Section II – Thyroid Diagnostic Testing

Thyroid Ultrasound

Thyroid ultrasound is the least invasive way to evaluate a solitary nodule in the thyroid. It simply involves having the patient lie comfortably with transmission gel on his or her throat as a sound-emitting instrument is gently massaged across the thyroid gland. How the emitted sound bounces back to the instrument can provide information to determine if the nodule has characteristics of a benign or malignant tissue. If the nodule has benign characteristics, a follow-up may be conducted in six months to a year to evaluate the gland again. If the nodule has malignant characteristics, a FNA biopsy or radioactive iodine uptake (RAIU) test may be conducted.

Malignant Characteristics of Thyroid Ultrasound

- Irregular border
- Hypoechoic (less echogenic than the surrounding tissue)
- Microcalcifications
- Taller-than-wide shape on transverse study
- Significant intranodular blood flow by power Doppler

Benign Characteristics of Thyroid Ultrasound

- Smooth borders
- Hyperechoic
- "Comet tail" artifact as sound waves bounce off intranodular colloid

Radioactive Iodine Uptake (RAIU)

An RAIU scan is a test conducted to evaluate the metabolic activity of iodine uptake by the thyroid gland. The test involves either oral ingestion of radioactive iodine or intravenous injection, followed by a radioiodine scan. "Most often, scanning is done 24 hours later if the radioisotope is given orally. If it is given intravenously, the scan is performed approximately 20 minutes later. For a thyroid scan, the patient is positioned lying down on his or her back, with the head tilted back. The radionuclide scanner, also called a gamma camera, is positioned above the thyroid area as it scans. The scanning takes 30-60 minutes."

"A normal scan will show a thyroid of normal size, shape, and position. The amount of radionuclide uptake by the thyroid will be normal according to established laboratory figures. In a normal scan there will be no areas where radionuclide uptake is increased or decreased. An area of increased radionuclide uptake may be called a hot nodule, or 'hot spot.' This means that the tissue is overactive." If there is a diffuse hot spot, it indicates that the entire gland is overactive, which is typically found with Graves' disease. If there is a localized hot spot, it indicates the presence of a toxic adenoma, which may need to be treated with surgery. Areas of "decreased radionuclide uptake may be called a cold nodule, or 'cold spot.' This indicates either" an area of decreased uptake that is not active in utilizing iodine, or that is not thyroid tissue. A localized cold spot strongly suggests the possibility of thyroid cancer. Cancer cells do not take up iodine. If the entire thyroid gland demonstrates low radioactive iodine uptake, it suggests thyroiditis. The RAIU test is typically not used to establish a clear diagnosis as an isolated test. It must be used with thyroid lab tests. It is typically a test conducted with hyperthyroid patients to help differentiate if the mechanism for thyroid hyperfunction is Graves' disease, toxic adenoma, or thyroiditis. The thyroid-stimulating immunoglobulins (TSI) test is currently the standard test to rule in or out the diagnosis of Graves' disease and RAIU. The RAIU test is commonly used in hyperthyroidism if the diagnosis of Graves' cannot be made due to negative TSI results. Listed below is the summary of potential diagnoses with uptakes that are increased, decreased, localized, and/or diffuse:

Diffusely High RAIU With Hyperthyroidism

- Thyrotoxicosis (Graves' disease)

Localized High RAIU With Hyperthyroidism

- Hyperfunctioning thyroid nodule (toxic adenoma)

Diffusely Low RAIU With Hyperthyroidism

- Subacute thyroiditis
- Postpartum thyroiditis (5%-8% of incidence)
- Painless lymphocytic thyroiditis
- Medications
  - Exogenous thyroid hormone intake
  - Excessive exogenous iodine intake
  - Goitrogenic medications (see previous list under "Goiter")

Localized Low RAIU With Hypothyroidism

- Requires biopsy for thyroid cancer evaluation

Fine-Needle Aspiration Biopsy

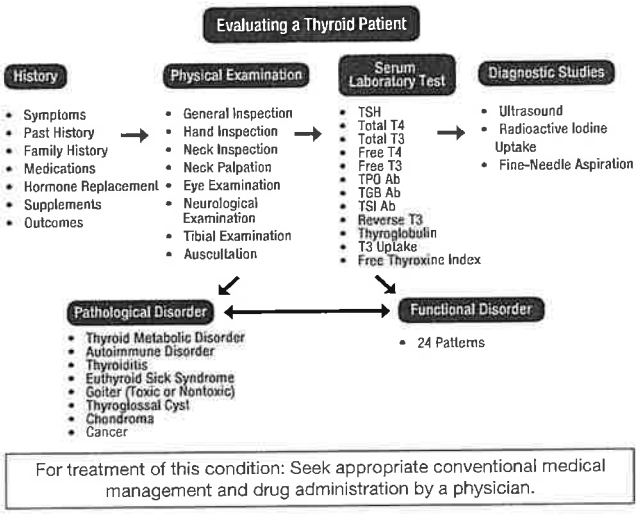
Fine-needle aspiration biopsy is the method of choice to identify if a thyroid mass is malignant or benign. It can also be used at times to identify the pathophysiology of the mass from its histological characteristics. This procedure is relatively brief, safe and accurate. It takes about 20 minutes to complete the procedure. Complications are bruising, bleeding, or infection. Needle aspiration does not require lacerations of tissue, as with surgery, and does not leave any scars. The needle-aspiration (suctions tissue) sampling is typically guided by ultrasound to increase accuracy and reliability. Usually anesthesia is not needed because the needle is very thin. A stinging sensation may be noted during needle insertion. After the sample is taken, pressure is applied to the biopsy site to stop the bleeding, and the site is covered with a bandage.

An FNA biopsy samples a thyroid nodule to help the physician determine if it is cancer or not. During the procedure, a small needle is passed through the skin into the thyroid nodule, and suction is applied to the syringe as the needle is moved up and down within the nodule. The process draws a sample of cells and other material up through the needle into the syringe. The needle is then removed, and the sample is placed on a glass slide where it can be stained and examined under a microscope.

Most thyroid samples are made up of two distinct components, epithelial cells and colloid cells. Epithelial cells are the predominant cells present in the tissue of the thyroid gland. Abnormalities in the formation of these cells can help determine the characteristics of the thyroid mass. In general, there are four diagnostic categories for samples from FNA biopsy, which are listed below.

Four Diagnostic Categories from FNA Biopsy

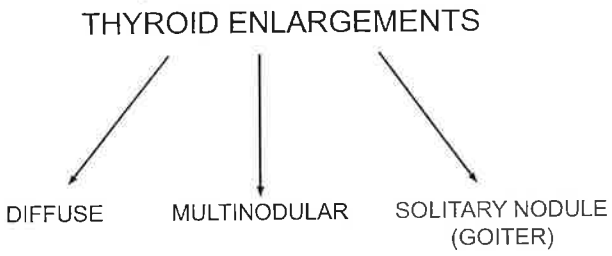
1. **Benign** – indicating that a sample is consistent with a noncancerous thyroid nodule.
2. **Malignant** – indicating that a sample is consistent with a cancerous thyroid nodule.
3. **Suspicious or Indeterminate** – indicating that a sample may either be benign or malignant. Nodules with such findings cannot be reliably diagnosed by FNA biopsy alone. Most (80%) are benign, but a minority (20%) are malignant.
4. **Inadequate** – indicating that there is not enough material in a sample to draw accurate conclusions.



Goiters, Nodules, and Masses

Thyroid enlargements can be classified into three main groups based on the anatomy of their enlargement: diffuse, multinodular, or solitary mass. Diffuse enlargement of the thyroid is defined as an enlargement of the entire gland, and may be related to hypothyroidism, iodine deficiency, Hashimoto's, Graves' disease, thyroiditis, anaplastic thyroid carcinoma, or any mechanism that promotes elevation of TSH. A multinodular enlargement is defined as multiple nodules caused by multinodular goiter. A single solitary nodule may be due to solitary nodular goiter, colloid cyst, adenoma, thyroid carcinoma, or lymphoma.

Three Classifications of Enlargements



Mechanisms for Thyroid Enlargements

Diffuse Enlargement

- Hypothyroidism
- Hashimoto's
- Iodine deficiency
- Graves' disease
- Thyroiditis
- Anaplastic carcinoma
- Pregnancy

Multinodular Enlargement

- Multinodular goiter

Solitary Nodule Enlargement

- Solitary nodular goiter
- Colloid cyst
- Adenoma
- Carcinoma
- Lymphoma

Diffuse Thyroid Enlargements

Goiter

Goiter is a diffuse enlargement of the thyroid gland. It is termed as toxic or nontoxic, depending on the resultant changes in thyroid hormone levels. Nontoxic goiter is a simple enlargement of the thyroid gland without any increased secretion of thyroid hormones. Nontoxic goiter can be endemic, most commonly due to a regional deficiency of iodine, or it can be sporadic. Sporadic goiter is typically the result of numerous etiologies, including pregnancy, Hashimoto's, goitrogen-induced, drug-induced, environmentally induced, or any mechanism that may raise TSH levels. Goiter may also be a result of thyroiditis, promoting inflammatory enlargement of the gland. Pregnancy may also promote nontoxic goiter because elevated human chorionic gonadotropin (hCG) can promote enlargement of the thyroid gland. Most clinical cases of nontoxic goiter today are sporadic and are typically not caused by simple iodine deficiency, but rather by dysfunction in thyroid physiology leading to increased stimulation of TSH. Chronic TSH stimulation promotes enlargement of the thyroid.

On the other hand, toxic goiter is an enlargement of the thyroid gland that produces pathologically excess amounts of thyroid hormones. Toxic goiter occurs in three forms, namely Graves' disease, toxic adenoma, and toxic nodular goiter. Toxic goiter leads to disorders promoted by increased basal metabolic rate from excess thyroid hormone release, such as cardiac arrhythmias, loss of weight, weakness, hyperkinesia, etc.

Once a diffuse enlargement of the thyroid gland is noted (goiter), simple observations in the patient's concomitant symptoms can help differentiate between toxic and nontoxic goiter. If a patient with a goiter has symptoms associated with increased basal metabolic activity, then toxic goiter is likely. If the patient has symptoms associated with low basal metabolic activity, then nontoxic goiter is likely. It should be noted that many patients with nontoxic goiter will not present with any thyroid symptoms. Follow-up laboratory testing to assess TSH, T4, T3, and thyroid antibodies should be conducted to further evaluate the goiter. Additionally, thyroid ultrasound, RAIU scan, or FNA biopsy may

be conducted to confirm that the goiter is not a result of malignant tissue growth. Thyroid cancer typically presents as a solitary nodule growth and not a diffuse enlargement. The exception to this is anaplastic thyroid cancer, but they account for only 1% of thyroid cancers and are most often found in the elderly. Typically, multinodular (not solitary) or diffuse enlargements of the thyroid are not due to malignancy. The incidence of malignancy within a multinodular goiter is usually significantly less than 5%.

Nontoxic Goiter

"The term nontoxic goiter refers to enlargement of the thyroid that is not associated with overproduction of thyroid hormones or malignancy. The thyroid becomes enlarged due" to chronic stimulation of TSH. Although the most common cause is iodine deficiency worldwide, it is not the most common cause in the United States. Goiter development in the U.S. is usually due to a defect in normal thyroid synthesis.

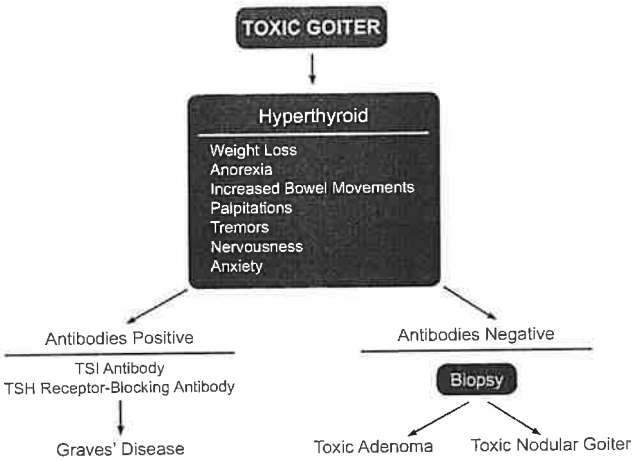
If the goiter becomes very large it may produce compressive symptoms in surrounding tissue such as the trachea and esophagus. Symptoms include coughing, waking up from sleep with feelings of difficulty breathing, and the sensation that food is getting stuck in the upper throat.

Nontoxic goiter may be treated if compressive symptoms occur or for cosmetic reasons. One should attempt to find the mechanism for the goiter if TSH is elevated—such as iodine deficiency, drug thyroid interaction, etc. In most instances, conventional endocrinologists will supply thyroid hormones to stabilize the size of the goiter. Thyroid hormones often will not cause the size of the goiter to decrease but it will usually keep it from growing any larger. If the mechanism for the goiter cannot be identified and thyroid hormones are ineffective to reduce the size of the goiter, surgery may be recommended.

Toxic Goiter

"In toxic goiter, the thyroid gland is diffusely hyperplastic and excessively overproduces thyroid hormones. This results in accelerated metabolism in most body tissues." Toxic goiter occurs in three forms, namely Graves' disease, toxic adenoma, and toxic nodular goiter. The most common source (85%) is Graves' disease, which can easily be identified by TSH antibodies/ immunoglobulins (TSI). In some cases of Graves' disease, there are genetic hypersensitivity reactions to an autoantibody of IgG type class that acts on the receptors for TSH. This leads to enlargement of the thyroid epithelium and the development of toxic goiter. (In nontoxic goiter, TSH stimulation promotes enlargement. In Graves' toxic goiter, the TSH-antibody promotes thyroid tissue enlargement.) If an autoimmune mechanism is not the etiology of toxic goiter, then toxic adenoma or toxic nodular tissue is autonomously promoting thyroid hormone secretion. Anatomically, they present as an isolated nodule rather than a diffuse enlargement of the thyroid. Autonomous hyperactivity is conferred by somatic mutations of TSH hormones or TSH receptors in a majority of toxic adenomas and toxic nodules. Toxic nodular and adenomatous goiter appear to develop more frequently in areas of endemic iodine deficiency. Iodine deficiency promotes thyroid cell hyperplasia and increases thyroid cell replication, which then leads to a greater predisposition of the development of TSH hormone and receptor mutations leading to toxic nodule. However, once toxic nodule or adenoma develop, iodine supplementation may promote greater thyroid hyperactivity (Jod-Basedow effect).

The typical patient with toxic goiter presents with both hyperthyroid symptoms and symptoms of obstruction from goiter. Obstructive goiter symptoms include dysphagia, dyspnea, or hoarseness. Excess thyroid hormone (thyrotoxicosis) symptoms include weight loss (muscle and fat), anorexia, increased bowel movements, palpitations, tremors, nervousness, and anxiety. Physical examination may reveal a goiter with or without a nodule, bruit over the thyroid, proximal muscle weakness, exaggerated deep tendon reflexes, and moist skin. Laboratory testing will demonstrate elevated T4 and/or T3 with a very suppressed TSH. Typical management of toxic goiter includes anti-thyroid medications, partial or complete thyroidectomy, and/or radioactive iodine, depending on the patient's age and clinical status of the gland.



For treatment of this condition: Seek appropriate conventional medical management and drug administration by a physician.

Goitrogens

Goitrogens are substances (drugs, chemicals, and foods) that disrupt the production of thyroid hormones by interfering with iodine uptake of the thyroid gland. This triggers the pituitary to release TSH that then promotes thyroid tissue growth leading to goiter. Although iodine itself is important for thyroid function, *it only takes a pinhead of iodine a day and only a teaspoon of iodine over a lifetime to avoid thyroid goiter* (that is not to say that individuals would not have better function with more iodine). Therefore, jumping to iodine need exclusively in cases of goiter in the U.S. population should not be the only factor considered. It is important to remember that any mechanism that leads to increased TSH production will promote goiter development. As far as goitrogenic mechanisms are considered, the impact of medications and environmental factors should be considered with the thyroid, especially after common mechanisms such as Hashimoto's, thyroiditis, and pregnancy have been considered.

Medications That May Promote Goiter

- Antithyroid medications (methimazole, Tapazole®, propylthiouracil): used for hyperthyroid patients
- Sulfonamides: used to prevent the growth of bacteria in the body
- Amiodarone (Cordarone®): used as an antiarrhythmic agent
- Ethionamide (Trecator®): used as an antibiotic for the treatment of tuberculosis
- Aminosalicylate Sodium: used an anti-infective for the treatment of tuberculosis
- Lithium: used for bipolar disorders, depression, anxiety, cluster headaches, and migraines
- Aminoglutethimide: used for the treatment of Cushing's syndrome and other endocrine disorders producing excess hormones (glucocorticoids, mineralocorticoids, estrogen, androgens)
- Salofalk®: used for the treatment of ulcerative colitis

Environmental Compounds That May Promote Goiter

- Mercury, arsenic, and potentially other heavy metals
- Nitrates
- Pesticide compounds
- Plastic compounds
- Ions such as thiocyanate and perchlorate
- Fluoride[179-180]

Foods That Promote Goiter

- Soybeans (and soybean products such as tofu)
- Pine nuts
- Peanuts
- Millet
- Strawberries
- Pears
- Peaches
- Spinach
- Bamboo shoots
- Radishes
- Horseradish
- Vegetables in the *Brassica* family
  - Bok choy
  - Broccoli
  - Brussels sprouts
  - Cabbage
  - Canola
  - Cauliflower
  - Collard greens
  - Kale
  - Kohlrabi
  - Mizuno
  - Mustard greens
  - Rapeseed
  - Rapini

Please see registered trademark ownership in Appendix 1.

- Rutabagas
- Tatsoi turnips

Cooking significantly inactivates the goitrogenic compounds found in food. "Both isoflavones (found in soy foods) and isothiocyanates (found in cruciferous vegetables) are heat sensitive, and cooking lowers their goitrogenic activity."[181-182]

"Some foods and drinks have an opposite effect on the thyroid gland; that is, they stimulate thyroid function rather than suppress it (examples include avocado, coconut, and saturated fat)."[183]

Clinical Applications of Goitrogens

There are two important considerations concerning goiters and goitrogens. First, once a goiter develops from a mechanism (iodine deficiency, environmental, Hashimoto's, etc), there is typically permanent hypertrophy of the gland. The gland anatomy (goiter) will typically not normalize once the goitrogenic source is removed or addressed. Therefore, *using goiter size as a clinical marker for effective treatment is limited.* Second, if there is an actual and active goitrogenic process blocking iodine uptake or iodine deficiency leading to goiter, the TSH must be elevated. In reality, we have numerous known and unknown goitrogenic exposures from diet and environment to our thyroid at any given time. It does not mean that goitrogenic compounds shut down our thyroid. In most instances, we have greater thyroid hormone-promoting mechanisms to counteract normal ongoing goitrogenic activity, leading to normal TSH and thyroid hormones. *Therefore, clinically pursuing goitrogenic models without an elevated TSH is not clinically efficient.*

The practice of having patients avoid goitrogenic foods in an attempt to improve thyroid function with a normal TSH is clinically naive. If goitrogenic activity is impacting thyroid synthesis to the degree that it disrupts thyroid hormone synthesis, the TSH must be elevated (functionally greater than 3.0). Foods that contain some goitrogenic activity are rich in flavonoids, antioxidants, fiber, minerals, and vitamins. They should not be avoided unless there is an identified clinical mechanism specific to goitrogenic or iodine uptake activity. In reality, these mechanisms are not very common in the U.S. population. However, in areas in the world with endemic regional iodine deficiency, goitrogenic compounds are a serious clinical consideration, or in areas of the world where goitrogenic foods are consumed as a major part of the diet, such as cassava in Africa. Although it's not common, U.S. population patients who consume abnormally excessive soy compounds or consume such things as cabbage juice on a regular basis may be at risk for goitrogenic activity.

Lastly, the use of goitrogenic compounds naturally (soy isoflavone supplements, cabbage juice, etc) to manage hyperthyroidism must be discussed. It is typically not a good idea to use goitrogenic compounds or avoid iodine foods to manage hyperthyroid activity. This strategy may backfire. Iodine restriction and heavy use of goitrogenic compounds will cause the thyroid to potentially become iodine deficient. Iodine deficiency promotes thyroid cell hyperplasia and increases thyroid cell replication. This then leads to a greater predisposition to the development of TSH hormone and receptor mutations leading to toxic nodule. However once toxic nodule or adenoma develop, iodine supplementation may promote greater thyroid hyperactivity (Jod-Basedow effect).

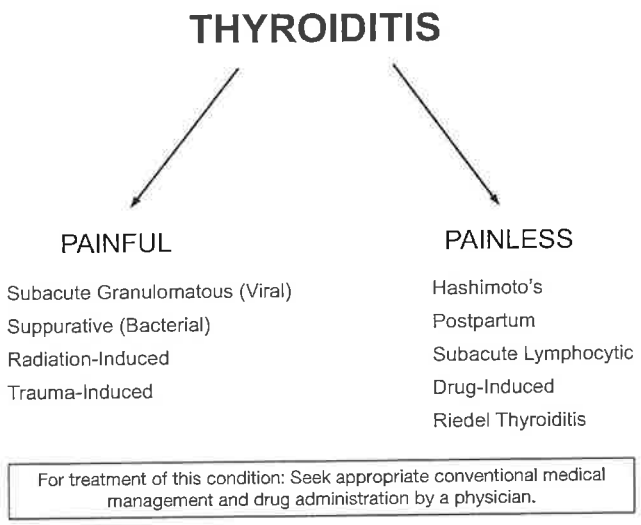
Thyroiditis

Thyroiditis is inflammation of the thyroid gland resulting in inflammation and damage to thyroid cells. Inflammation of the thyroid gland may be present with normal, elevated, or depressed thyroid function. Thyroiditis is considered a disorder of the immune system, not a primary endocrine disorder. Most types of thyroiditis are three to five times more likely to be found in women than in men. The average age of onset is between thirty to fifty years of age. The condition tends to be geographical and seasonal, and is most common in summer and fall. There are many forms of thyroiditis and each one has its own causes, clinical features, diagnosis, duration, resolutions, conditions, and risks. The various mechanisms of thyroiditis include: Hashimoto's thyroiditis (autoimmune), drug-induced thyroiditis (amiodarone, lithium, or interferon), radiation-induced thyroiditis, and antigen-induced thyroiditis (viral or bacterial). However, the most common type is Hashimoto's thyroiditis.

Thyroiditis has a diffuse set of symptoms, none of which are exclusive to the condition; therefore, it is difficult to diagnose based solely on symptomatology. The diagnosis of thyroiditis is typically investigated after an enlarged thyroid gland is noted during thyroid palpation. The patient's symptoms, such as hyperthyroid, hypothyroid, or pain, will help establish the type of thyroiditis in conjunction with laboratory testing (ESR,

thyroglobulin, thyroid antibodies, TSH, T4, T3, white blood cells [WBCs]), radiodine uptake (depressed uptake in thyroiditis), ultrasound, and FNA biopsy (to find out what is attacking the thyroid) if necessary.

The initial differential diagnosis criteria to help differentiate thyroiditis is pain. Painful thyroiditis is associated with four possibilities. These include subacute granulomatous thyroiditis, suppurative thyroiditis, radiation, or trauma-induced thyroiditis. Painless thyroiditis is associated with four other possibilities. These include Hashimoto's, postpartum, subacute lymphocytic, and drug-induced thyroiditis.



Painful Thyroiditis Conditions

Subacute Granulomatous Thyroiditis: also known as "giant cell thyroiditis," "subacute thyroiditis," or "de Quervain's thyroiditis," and is attributed to a thyroid infection. It is the most common cause of thyroid pain. This type of thyroiditis occurs equally in both men and women. It usually starts out as a harmless viral illness such as a flu or cold, which invades the thyroid gland, causing thyroiditis. The inflammation leads to a painful throat. It may affect one lobe or the entire thyroid. "The summer peak incidence of thyroiditis coincides with the peak incidences of Coxsackie virus groups A and B and echovirus."[184]

"Symptoms and signs of subacute granulomatous thyroiditis include a prodrome of myalgias, pharyngitis, low-grade fever, and fatigue, followed by a tender, diffuse goiter and neck pain that often radiates up to the ear. As the disease progresses, there may be a 'march' of tenderness across the gland, with new parts of the a thyroid gland becoming painful and tender." During the inflammatory response, it is possible for a hyperthyroid state to be generated from severe destruction of thyroid cells leading to abnormal release of thyroid hormones into the bloodstream. This occurs in 50% of subacute thyroiditis cases. Subacute thyroiditis typically lasts three to six weeks. A triphasic response may occur in subacute thyroiditis that includes hyperthyroidism the first few weeks, hypothyroidism for up to six months, followed by possible return to euthyroidism within six to 12 months. Approximately 10%-15% of patients will develop lifelong hypothyroidism from extensive thyroid tissue destruction and will require thyroid hormone replacement. These individuals may also develop atrophy and shrinking of their thyroid gland that is most noticeable when only one side of their gland is involved. (Please refer to the American Academy of Family Physicians [www.aafp.org/afp/2006/0515/p1769.html] for more information.)

The diagnosis of subacute thyroiditis is typically done with a history of pain in the thyroid or throat. Laboratory testing may demonstrate elevated ESR (erythrocyte sedimentation rate), elevated thyroglobulin, elevated total WBCs, elevated lymphocytes, anemia of chronic disease (microcytic hypochromic anemia with normal ferritin and TIBC), and elevated CRP (C-reactive protein). The TSH may be elevated, depressed, or normal, depending on the phase of the thyroiditis. Thyroid antibodies are negative in viral subacute thyroiditis, unless there is overlapping Hashimoto's or autoimmune thyroid response. A radioactive iodine scan will show almost no concentration of radioiodine by the thyroid gland because the cells are temporarily injured during the inflammatory response. The conventional treatment for thyroiditis typically involves aspirin or ibuprofen to decrease inflammation and pain. If no improvement occurs within a week, prednisone may be given in a dose of 40-60 mg daily, tapered to complete discontinuation over four to six weeks. If the thyroid is severely destroyed after the subacute process, thyroid hormone replacement therapy may be provided.

Suppurative Thyroiditis: an extremely rare form of thyroiditis caused by bacterial infection (especially *Streptococcus pyrogens, Streptococcus aureus, or Streptococcus pneumoniae*). Even more rare, but a possible mechanism of infection may be fungal, mycoplasmal, or parasitic. Suppurative thyroiditis is rare because antibiotics are commonly used and bacterial infections are treated before they spread to the thyroid

gland. Suppurative thyroiditis is also rare because the thyroid gland is generally resistant to infection because of its rich blood supply, lymphatic drainage, high iodine and hydrogen peroxide content, and encapsulation. The infection causes pus to collect and form an abscess in the thyroid gland. This may present clinically as an acute unilateral painful mass in the neck with erythema of the skin overlying the exquisitely tender thyroid. Fever, dysphagia, and dysphonia also are present. Laboratory testing typically demonstrates elevated ESR, CRP, WBCs, and thyroglobulin. The TSH may be normal, depressed, or elevated depending upon the degree of thyroid destruction produced by the infection. Typically, the TSH is normal. Thyroid antibodies are normal unless there are overlapping autoimmune reactions or Hashimoto's. A radioactive iodine scan will show almost no concentration of radioiodine by the thyroid gland because the cells are temporarily injured during the inflammatory response. Fine-needle aspiration of the lesion with Gram stain in culture is the most useful diagnostic test. Conventional treatment is parenteral (IV) antibiotics and possible surgical drainage of the infected thyroid abscess.

Radiation-Induced Thyroiditis: typically seen in hyperthyroid patients who are being treated by radioactive iodine therapy. Radioactive iodine therapy is used to destroy overactive thyroid cells in the management of hyperthyroidism; however, 1% of patients develop radiation thyroiditis between five and 10 days after the therapy. It is usually caused by too much radioactive substance, leading to rapid destruction of the thyroid parenchyma, resulting in further inflammation of the thyroid gland. It is important to remember that hyperthyroidism from an autoimmune mechanism is painless; therefore, if pain develops after radioactive iodine therapy, there is radiation-induced thyroiditis. The gland eventually undergoes extensive fibrosis in approximately 6-18 weeks, and the person may develop lifelong hypothyroidism from the destructive process. Conventional treatment for radiation-induced thyroiditis includes NSAIDs, prednisone, and beta-blockers to minimize the impact of increased thyroid hormone influences on peripheral basal metabolic rate. The most important clinical differential diagnostic factor for this pattern is the development of pain in the thyroid and increased hyperthyroid symptoms between five and 10 days after therapy. Radiation-induced thyroiditis may also develop from radiation therapy used to treat lymphoma of the head and neck. The major risk factors for developing thyroid damage after external irradiation are high-dose irradiation, younger age, female gender, and preexisting hypothyroidism.[185-186]

Trauma-Induced Thyroiditis: a rare condition, but may occur with significant trauma to the thyroid gland—such as being punched in the neck, being hit by a baseball to the neck, having a seatbelt restrain the body against the thyroid during a car accident, etc. The thyroiditis is self-limiting and the history, with obvious signs of bruising, allows the diagnosis to be made.

Painless Thyroid Conditions

Hashimoto's Thyroiditis: is the most common type of thyroiditis, and it is painless.[187] It is also called chronic lymphocytic thyroiditis. It is caused by an autoimmune response, and in the majority of cases there is a TH-1–dominant immune response. Most patients with Hashimoto's thyroiditis typically do not realize that they even have the disorder since the pathophysiology of the disease involves a slow and gradual destruction of the thyroid. The person typically develops gradual symptoms of fatigue, depression, weight gain, and constipation. During the development of the disease process, the patient may have totally normal TSH, T3, and T4 levels until enough thyroid tissue is lost, leading to the inability of the thyroid gland to meet the body's demands, resulting in elevation of TSH. Additionally, the TSH fluctuates routinely in the Hashimoto's process due to relapsing and remitting flare-ups of the immune attack against the thyroid. In a majority of cases with an ongoing chronic thyroiditis the thyroid panel (TSH, T4, and T3) are normal.[3] Although the patient may have lost ideal thyroid hormone output from destruction leading to elevated TSH, at the time the TSH assay was conducted a recent relapse of the attack may cause an increase of thyroid hormones in the blood stream, thus making TSH look normal. The best measurements to identify Hashimoto's thyroiditis are thyroid peroxidase antibodies (TPO Ab) and anti-Tg Ab. TPO antibodies are present in 90%-95% cases, whereas antithyroid globulin antibodies are present in only 20%-50% of cases.[188] It should be noted that thyroid antibodies may fluctuate; therefore, testing them more than once may be necessary. If they are positive, Hashimoto's thyroiditis is likely and thyroid failure will most likely occur at some point in their lifespan.[3] Occasionally, if the immune response in Hashimoto's is abrupt and severe, thyroid hyperfunction can develop.

If Hashimoto's is not managed and hypothyroid

develops, increased levels of TSH may promote hypertrophy of the thyroid and cause goiter. Around 90% of Hashimoto's patients have a symmetrical, diffusely enlarged gland with a firm pebbly texture, whereas around 10 percent have thyroid atrophy.[187] Hashimoto's thyroiditis is the number one cause of hypothyroidism and goiter in the United States.[1-2] During the chronic and progressive state of Hashimoto's, the only laboratory marker present is typically the thyroid antibodies. If the Hashimoto's thyroiditis develops into a hyperthyroid response due to abrupt immune attack, it may present identically to thyroid hyperthyroidism (depressed TSH, elevated thyroid hormones, and positive thyroid antibodies). Fine-needle aspiration biopsy is needed to differentiate Hashimoto's from Graves' disease and other thyroid disorders leading to hyperfunction (toxic nodule, anaplastic cancer, etc). Radiodine uptake may be low, normal, or high, depending on the aftermath of the immune response on the thyroid and is not used for diagnosis.

Typically, most patients who have chronic thyroiditis (Hashimoto's) are not identified in conventional healthcare until they progress to frank hypothyroidism (elevated TSH) or present with diffuse goiter. This is unfortunate because these patients suffer from hypothyroid symptoms along the way. This is partly due to the fact that the cytokines (TNF-alpha, IL-6, interferon, etc) elevated in the autoimmune response in Hashimoto's prevents the thyroid receptor proteomic expression, leading to decreased thyroid hormone tissue responses.[20-22]

Additionally, associated disorders and risks accompany Hashimoto's. First, the condition is typically found along with other autoimmune disorders—such as pernicious anemia, celiac disease, autoimmune diabetes, Addison's disease, etc.[1,187] Secondly, the appearance of a rapidly growing nodule, not the typical diffuse enlargement (goiter), should raise suspicion of primary thyroid lymphoma, because this is 60%-80% more likely in patients with Hashimoto's than in the general population.[189] Hashimoto's is also associated, although less strongly, with papillary carcinoma.[190] An FNA biopsy should be done in any rapidly growing nodule to identify the cause.

Postpartum Thyroiditis: occurs in approximately 5%-7% of women within two to four months after delivery. It is the result of an autoimmune process. It is a common etiology for postpartum depression. During the third trimester of pregnancy, a female is TH-2 dominant, and postpartum they become TH-1 dominant. These immunological polar shifts in combination with certain genotypes may lead to the development of postpartum thyroiditis. It presents as a painless, small, nontender, firm goiter. Typically, the patient notices an enlarged thyroid and/or symptoms of hyperthyroidism that include increased anxiety, restlessness, insomnia, weight loss, and difficulty concentrating. Hypothyroidism occurs between four to eight months after delivery, with a duration of nine to 12 months and with recovery taking place at 12 months after onset. Most patients (80%) have normal thyroid function at one year.[191] However, 30%-50% percent of patients develop permanent hypothyroidism.[192-193] Elevated levels of thyroid antibodies are found in 80% of patients, but the erythrocyte sedimentation rate is typically normal. It is important to distinguish painless thyroiditis from Graves' disease occurring in the postpartum period. The presence of an audible bruit over the gland, exophthalmos, hypervascularity with increased blood flow seen on Doppler ultrasonography, thyroid-stimulating immunoglobulins in the serum, and high RAIU are characteristic of Graves' disease, but not postpartum thyroiditis. Thyroid uptake and scan should not be performed in women who are breastfeeding.

It should be noted that women with euthyroidism who have TPO antibodies have a 25% risk for developing postpartum thyroiditis.[192] Therefore, thyroid antibodies should be considered in the prenatal evaluation. Additionally, patients with postpartum thyroiditis who have elevated thyroid antibodies have a 70% risk of recurrence following a subsequent pregnancy.[194]

Subacute Lymphocytic Thyroiditis: "also known as silent sporadic thyroiditis or painless sporadic thyroiditis." It is of an autoimmune origin and usually presents with a small goiter (50% of cases), hyperthyroid responses, and is painless. In 50% of the cases, the thyroid antibodies are positive. It clinically mimics a hyperthyroid Hashimoto's thyroiditis, especially when antibodies are present, but the histopathology of the condition is different. In subacute lymphocytic thyroiditis, no fibrosis occurs. The condition can only be differentially diagnosed from Hashimoto's thyroiditis that is active by FNA biopsy. The condition is very rare compared to Hashimoto's thyroiditis. "It is differentiated from subacute thyroiditis (not lymphocytic) by the absence of thyroid pain and tenderness. It is differentiated from Graves' disease by the lack of

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thyroid thrill or bruit, no ophthalmopathy, no pretibial myxedema, no thyroid stimulating antibodies/ immunoglobulins (TSI) elevations, and by a low or absent rather than elevated RAIU," and by FNA biopsy. The hyperthyroid stage averages 3-4 months. Eventually, these patients will develop hypothyroidism from a less-active autoimmune response in subsequent years. Interestingly, four times more women are affected than men, and the risk is increased in persons who live in areas of iodine deficiency.[188,194]

Drug-Induced Thyroiditis: Certain medications in the presence of autoimmune thyroid physiology may promote thyroiditis due to their stimulatory impact on up-regulating an autoimmune response. "*Amiodarone (Cordarone®)* is an antiarrhythmic agent used for various types of tachyarrhythmias, both ventricular and supraventricular (atrial) arrhythmias. It is used in arrhythmias that are otherwise difficult to treat with medication." Amiodarone is rich in iodine content (37.3% by weight) and resembles thyroid hormones with its ability to bind to nuclear thyroid receptors. It may promote increased thyroid activity by increasing the thyroid nuclear response, and it may also feed a source of iodine to an autoimmune hyperactive response.

Interferon (Infergen®; Intron® A, Roferon®-A combination therapy) and *interleukin-2* are typically used to simulate the immune system in cases such as hepatitis C, cancer, HIV, etc. Unfortunately, it may promote an increased TH-1 cell-mediated response and increase the immune attack on the thyroid. The majority of autoimmune Hashimoto's are TH-1 dominant.

Long-term use of lithium for manic-depression and depression has been shown to promote both hypothyroidism and, less commonly, hyperthyroidism. Lithium may act as a goitrogen and block iodine uptake leading to hypothyroidism. However, lithium's blocking of iodine uptake may increase the metabolic activity of the thyroid gland and promote a hyperactive thyroiditis in the presence of a preexisting autoimmune thyroid disorder. These drugs may cause destructive thyroiditis in the presence of thyroid autoimmune diseases, as evidenced by elevated thyroid antibodies. It should be noted "that the thyroid abnormalities usually resolve with discontinuation of the responsible drug." This is another reason that thyroid antibodies should be used as a general screening marker for an annual examination.[195-199]

Riedel Thyroiditis: a very rare condition and also known as fibrous thyroiditis. It is typically a systemic fibrotic process affecting multiple tissues. When it causes fibrotic thyroiditis responses, it has a characteristically unique presentation of a rock-hard, wood-like, fixed, painless goiter, often associated with symptoms of esophageal and tracheal compression. These symptoms include stridor, dyspnea, a suffocating feeling, dysphagia, and hoarseness. Treatment is surgical removal of the mass. Approximately 1/3 of patients have hypothyroidism from extensive scarring and are treated with thyroid hormone replacement.

Nodules

Multinodular Enlargement

- Multinodular goiter

Solitary Nodular Enlargement

- Solitary nodular goiter
- Colloid cyst
- Adenoma
- Carcinoma
- Lymphoma

Mechanisms for Thyroid Enlargements

Thyroid nodules are basically increased growths or lumps that arise in an otherwise normal thyroid gland. They are only typically noticed if they become very large or if the person is extremely thin. Many individuals have benign nodules even at young ages. As people age, they will develop more nodules. Most people will develop a nodule by the time they are 50 years old, and almost 10% of the adult population has a nodule. Autopsy studies reveal the presence of thyroid nodules in 50% of the population, so they are fairly common.

The biggest concern with a nodule is cancer, the second is if the nodule compresses surrounding structures, and the third is if the nodule produces hormones. In most instances, none of these concerns apply, and a nodule is nothing more than a simple structural lump in the tissue of the thyroid. In some instances, a nodule may develop just as a goiter develops due to chronic stimulation of TSH. Chronic TSH stimulation to the thyroid tissue enlarges the gland in the form of goiter and nodules. However, most nodules are not due to elevated TSH or any other dysfunctional mechanism.

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In any case, when a nodule is identified, further testing may be warranted. A basic thyroid panel should be conducted to see if the individual has elevated TSH or to evaluate if the nodule itself is producing thyroid hormones. If the nodule is producing hormones, the TSH will be suppressed, and if the nodule is being promoted by a metabolic thyroid hypofunction, the TSH will be elevated. Second, an ultrasound may be conducted to determine the characteristics of a nodule. Benign nodules typically have identifiable characteristics with ultrasound study. These include: nice sharp edges surrounding the nodule, nodules filled with fluid instead of tissue, lack of blood flow through it, and the presence of multiple nodules (multinodular goiter is almost always benign). Ultrasound characteristics of malignant mass include: tissue-filled mass with blood flow, non-demarcated borders, and a solitary mass. It is important to note that cancer cannot be diagnosed by ultrasound alone. The definitive test to diagnose if the nodule is cancerous or benign is FNA biopsy.

Common types of benign thyroid nodules are adenomas (overgrowths of normal thyroid tissue), thyroid cysts, and Hashimoto's thyroiditis.

Uncommon types of benign thyroid nodules are due to subacute thyroiditis, painless thyroiditis, unilateral lobe agenesis, or Riedel struma.

Facts About Thyroid Nodules

- One in 12 to 15 young women has a thyroid nodule.
- One in 40 young men has a thyroid nodule.
- More than 95% of all thyroid nodules are benign.
- Some are actually cysts which are filled with fluid rather than thyroid tissue.
- Most people will develop a thyroid nodule by the time they are 50 years old.
- The incidence of thyroid nodules increases with age:
  - 50% of 50 year olds will have at least one thyroid nodule
  - 60% of 60 year olds will have at least one thyroid nodule
  - 70% of 70 year olds will have at least one thyroid nodule

Source: Endocrineweb. http://www.endocrineweb.com/conditions/thyroid/thyroid-nodules

Three Questions for All Thyroid Nodules

1. "Is the nodule one of the few that are cancerous?"
2. "Is the nodule causing trouble by pressing on other structures in the neck?"
3. "Is the nodule making too much thyroid hormone?"

Thyroid cancers typically present as a dominant solitary thyroid nodule. However, 95% of solitary thyroid nodules are benign.

Masses

Thyroid Cancer

"There are about 20,000 new cases of papillary thyroid cancer each year in the United States."[200] "Females are more likely to have thyroid cancer at a ratio of 3:1. Thyroid cancer can occur in any age group, although it is most common after age 30, and its aggressiveness increases significantly in older patients."[201] There are four kinds of malignant thyroid cancers: papillary, follicular, medullary, or anaplastic. "Papillary and follicular tumors are the most common. They grow slowly and may recur, but are generally not fatal in patients under 45 years of age. Medullary tumors have a good prognosis if restricted to the thyroid gland and a poorer prognosis if metastasis occurs. Anaplastic tumors are fast-growing and respond poorly to therapy."[202] Distinction between these types of tumors can be made by FNA biopsy.

Percentage of Thyroid Cancer Types[201]

- Papillary and/or mixed papillary/follicular ~80%
- Follicular ~15%
- Medullary ~3%
- Anaplastic ~2%

Source: Endocrineweb. http://www.endocrineweb.com/conditions/thyroid/thyroid-nodules

Characteristics of Papillary Thyroid Cancer

- Most common of all thyroid cancers
- Peak onset ages 30-50
- Prognosis directly related to size (less than ½ inch has a good prognosis)
- Accounts for 85% of thyroid cancers due to radiation exposure
- Spread to lymph nodes of the neck present in more than 50% of cases
- Distant spread to lung or bones is very uncommon
- Overall cure rate very high (nearly 100% for small lesions in young patients)

Characteristics of Follicular Thyroid Cancer [200]

- "Peak onset ages 40-60
- Females more common than males by a 3:1 ratio

- Prognosis directly related to tumor size (less than 3/8 inch has a good prognosis)
- Rarely associated with radiation exposure
- Spread to lymph nodes is uncommon (10%)
- Invasion into vascular structures (veins and arteries) within the thyroid gland is common
- Distant spread to lungs and bones is uncommon, but more common than with papillary cancer
- Overall cure rate high (near 95% for small lesions in young patients), decreases with advanced age"

Characteristics of Medullary Thyroid Cancer

- Females more common than males (except for inherited cancers)
- Regional metastasis (spread to neck lymph nodes) occurs early
- Spread to distant organs occurs late and can be to the liver, bone, brain, and adrenal medulla
- Not associated with radiation exposure
- Poor prognostic factors include age older than 50, male, metastasis, and when seen in patients with other endocrine tumors due to MEN (multiple endocrine neoplasia)-II syndrome
- Residual disease (following surgery) or recurrence can be detected by measuring calcitonin every four months for the first few years, then every six months after that.

Characteristics of Anaplastic Thyroid Cancer [200]

- Peak onset age 65 or older
- Very rare in young patients
- Males more common than females by a 2:1 ratio
- Typically presents as a rapidly growing neck mass
- Can occur many years after radiation exposure
- Spread to lymph nodes of the neck present in more than 90% of cases
- Distant spread to lungs and bones is very common, even when first diagnosed
- Overall cure rate is very low

Most patients who have thyroid cancer do not present with thyroid symptoms. The only exception is a progressed thyroid cancer with metastasis to the thyroid gland, causing hyperthyroidism symptoms due to release of thyroid hormones. Hypothyroid symptoms are typically not associated with thyroid cancers. The majority of patients present with a nodule on their thyroid that typically does not cause symptoms. It is important to note that 99% of thyroid nodules are not cancer.

However, when a thyroid cancer does begin to grow within a thyroid gland, it almost always does so within a discrete nodule within the thyroid. Thyroid cancers are typically painless. Occasionally, symptoms such as hoarseness and neck pain may develop if surrounding tissues—such as the laryngeal nerves—are compressed or infiltrated. Enlarged cervical lymph nodes may also be present with thyroid cancer.

The typical case history and management of a thyroid cancer patient starts with identification of a neck mass, usually first noted by the patient. An ultrasound is conducted to confirm the presence of a mass. If the ultrasound findings are not clear, a radioactive iodine scan may be done to determine if the nodule is hot or cold. However, in many instances, an FNA biopsy is conducted after the mass is identified without the radioactive iodine scan. Fine-needle aspiration biopsy is considered the most cost-effective, sensitive, and accurate test. The biopsy is evaluated by a pathologist to determine if it is benign or cancerous. In many instances, the entire thyroid is removed (sometimes this is done during the same operation where the biopsy was done.) The lymph nodes in the neck will be assessed to see if they need to be removed also. In the case of anaplastic thyroid cancer, a decision will be made regarding the possibility of tracheotomy. About 4-6 weeks after the thyroid has been removed, the patient will undergo radioactive iodine treatment. This will require taking a single pill. The patient goes home, avoids contact with other people for a couple of days so they are not exposed to the radioactive materials. A week or two after the radioactive iodine treatment, the patient is placed on thyroid hormone replacement and monitored on a regular basis with serum TSH tests. Thyroid cancer management has high success rates. The success rate for the most common cancers (95% of thyroid cancers), papillary and follicular, are better than 97%. Medullary cancers (4%) are less successful because they tend to metastasize to the lymph nodes very early on. Anaplastic thyroid cancers (1%) have very poor prognosis and typically have very rapid and early spreading before the mass is identified.

Midline Masses and Thyroglossal Cysts

If a midline mass is found over the thyroid, it is first important to note if the mass moves up or down with swallowing. If the mass does not move up or down with swallowing, it is most likely a chondroma or a dermoid cyst. "A chondroma is a benign cartilaginous tumor that

is encapsulated within the lobular growing patterns of cartilage." It presents as a mass over the thyroid that does not move with swallowing. A dermoid cyst is a benign cystic teratoma that presents as a mass over the thyroid that does not move with swallowing.

If the mass does move with swallowing, it is then important to note if the mass moves when the tongue is protruded. If the mass moves with tongue protrusion, then it is most likely a thyroglossal cyst. "A thyroglossal cyst is a fibrous cyst that forms from persistent thyroglossal duct" during embryological development for a small portion of the population and presents as a midline neck lump. "Thyroglossal duct cysts most often present with a palpable asymptomatic midline neck mass below the level of the hyoid bone. The mass on the neck moves during swallowing. Diagnostically, a thyroglossal cyst will move upwards with protrusion of the tongue. Some patients will have neck or throat pain, or dysphagia. The persistent duct or sinus can promote oral secretion, which may cause cysts to become infected. Up to half of thyroglossal cysts are not diagnosed until adult life. The tract can lie dormant for years or even decades until some kind of stimulus leads to cystic dilation. Infection can sometimes cause the transient appearance of a mass or enlargement of the cyst, at times with periodic recurrences. Spontaneous drainage may also occur. Treatment for a thyroglossal cyst is surgical resection, often requiring concomitant removal of the midsection of the hyoid bone. Although generally benign, the cyst will be removed if the patient exhibits difficulty in breathing or swallowing, or if the cyst is infected. Even if the symptoms are not present the cyst may be removed to eliminate the chance of infection or development of a carcinoma, or for cosmetic reasons if there is unsightly protrusion from the neck."

If the mass does not move with tongue protrusion, then it is a thyroid enlargement and palpation of the thyroid can further subdivide the mass into a diffuse multinodular mass, or a solitary nodule. The assessment of nodules will be discussed next. It is important to note that nodules are not always midline and generally can be just off the midline.

For treatment of this condition: Seek appropriate conventional medical management and drug administration by a physician.

Where a condition is identified or a lab result is outside the normal range: Patient should seek appropriate conventional medical management and drug administration by a physician.

Case Examples

Case 9

- Neck Palpation: Goiter
- Ultrasound: Smooth Borders
- RAIU: Diffusely Low Uptake
- Fine-Needle Aspiration Biopsy: Normal

Case 10

- Neck Palpation: Single Nodule
- Ultrasound: Hypoechoic
- RAIU: Localized Low Uptake
- Fine-Needle Aspiration Biopsy: To Be Performed

Case 11

- Neck Palpation: Multinodular
- Ultrasound: To Be Performed
- RAIU: To Be Performed
- Fine-Needle Aspiration Biopsy: To Be Performed

Case 12

- Neck Palpation: Goiter With Single Nodule With Different Structure
- Ultrasound: Smooth Border
- RAIU: Localized High Uptake
- Fine-Needle Aspiration Biopsy: To Be Performed

Case 13

- Neck Palpation: Goiter
- Ultrasound: Microcalcifications
- RAIU: Diffusely Low Radioiodine Uptake
- Fine-Needle Aspiration Biopsy: Inflamed Tissue

Case 14

- Neck Palpation: Goiter
- Ultrasound: Significant Intranodular Blood Flow
- RAIU: Diffusely High
- Fine-Needle Aspiration Biopsy: To Be Performed

Case 15

- Neck Palpation: Painful and Sore Goiter
- Ultrasound: Smooth Borders, Hyperechoic
- RAIU: Diffusely Low Uptake
- Fine-Needle Aspiration Biopsy: Inflamed Tissue

Thyroid Disorders

Hypothyroidism

- Hypothyroidism is a condition in which the thyroid gland does not produce enough thyroid hormones. Thyroid hormones are responsible for activating the metabolic rate of all cells; therefore, the symptoms of hypothyroidism are diverse.
- The clinical symptoms associated with hypothyroidism include fatigue, depression, weight gain, lack of motivation, poor recovery from injuries and workouts, constipation, etc.
- The clinical signs include hair thinning; hair loss; swelling and puffy face; obesity; dry, brittle hair; dry skin; and slow deep-tendon reflexes.
- Women are affected by hypothyroidism more than men, especially as they age. The incidence increases with age.
- Hypothyroidism is identified with elevated serum thyroid TSH elevations.
- Several mechanisms are related to the development of hypothyroidism that can be divided into autoimmune and nonautoimmune mechanisms. The most common cause of hypothyroidism in the United States in Hashimoto's autoimmune disorder. The most common cause in undeveloped and malnourished countries is iodine deficiency.

Hashimoto's

- Hashimoto's is an autoimmune disease and is the most common cause of primary hypothyroidism in North America. It occurs far more often in women than men. It is 10-20 times more common in women.
- It can present as hypothyroidism, called Hashimoto's hypothyroidism, or it can present as hyperfunction, called Hashimoto's thyroiditis. The most common mechanism is a slow immune attack on the thyroid that gradually presents as clinical hypothyroidism. If an abrupt and aggressive immune attack develops, the result is hyperthyroidism, defined as Hashimoto's thyroiditis.
- It is identified with serum laboratory testing elevations of TPO Abs and/or Tg Abs.
- A small percentage of the population will not have antibodies present but will show histological characteristics of Hashimoto's.
- The clinical presentation of Hashimoto's hypothyroidism includes a mixture of hyper- and hypothyroid symptoms due to low-grade, relapsing, and remitting attacks on the thyroid gland.
- The HLA-DR5, HLA-DQ, and CTLA-4 genes have been associated with Hashimoto's.
- The current medical management of the condition is to provide replacement hormones and have serum TSH conducted regularly to monitor correct increase of thyroid replacement as the thyroid gland is continuously destroyed. No treatment or management is available or conducted for the autoimmune mechanisms.
- Most individuals with hypothyroidism have Hashimoto's hypothyroidism but are unaware of the proper diagnosis because thyroid antibodies are rarely conducted in the current healthcare model since their results would not change the outcome of their treatment.

Graves' Disease

- Graves' disease is an autoimmune disease of the thyroid gland leading to severe destruction of the thyroid gland and antibody production that stimulates thyroid hormone activity with resultant increase in circulating thyroid hormones.
- It affects up to 2% of the female population and has a female:male incidence of 5:1 to 10:1. It has a strong hereditary component.
- It typically causes the thyroid gland to enlarge and become overactive with related hyperthyroid symptoms, such as increased heartbeat, muscle weakness, disturbed sleep, and irritability.
- The clinical signs of Graves' include goiter, proptosis, exophthalmos, clubbing of fingers, restlessness, onycholysis, hyperpigmentation, muscle wasting, tremor of hands, atrial fibrillation, tachycardia, and increased deep tendon reflexes.
- The condition is diagnosed with elevations of serum laboratory thyroid receptor antibodies (TR Abs). It also presents with elevations of thyroid peroxidase antibodies (TPO Abs) and/or Tg Abs.
- Laboratory pattern reflects hyperthyroidism that presents as suppressed TSH and elevated thyroid hormones.
- Graves' disease typically presents as part of multiple endocrine tissues attacked in polyendocrine autoimmune syndrome and often accompanies the diagnosis of other autoimmune diseases such as rheumatoid arthritis, SLE, etc.

Euthyroid Sick Syndrome

- It is described as abnormal findings on thyroid function tests (T3, TSH, T4, free T4, free T3, TBG) that occur in the setting of an NTI without

preexisting hypothalamic-pituitary and thyroid-gland dysfunction.

- Proposed mechanisms for these imbalances include: inhibition or excess production of thyroid-binding proteins, cytokine dysregulation, deiodination, inhibition of thyroid-releasing hormone, and thyroid-stimulating hormone secretion.
- Changes of laboratory markers within the laboratory range may result from gastrointestinal diseases, pulmonary diseases, cardiovascular diseases, renal diseases, infiltrative and metabolic disorders, inflammatory conditions, myocardial infarction, starvation, sepsis, and burns.
- Changes of laboratory markers within functional ranges may result from active stress responses, inflammatory conditions, overtraining, elevated cortisol, elevated homocysteine, oral contraceptives, hormonal disorders, and chronic immune activation.

Subacute Granulomatous Thyroiditis

- Presents as a painful and swollen throat
- Symptoms and signs include a prodrome of myalgias, pharyngitis, low-grade fever, and fatigue, followed by a tender, diffuse goiter and neck pain that often radiates up to the ear.
- It usually starts out as a harmless viral illness such as flu or cold that invades the thyroid gland causing thyroiditis.
- Subacute thyroiditis typically lasts for three to six weeks. A triphasic response may occur in subacute thyroiditis, which includes hyperthyroidism the first few weeks, hypothyroidism for up to six months, followed by a possible return to euthyroidism within six to 12 months.
- Approximately 10%-15% of individuals will develop lifelong hypothyroidism from extensive thyroid tissue destruction and will require thyroid hormone replacement.
- Laboratory testing may demonstrate elevated ESR, elevated thyroglobulin, elevated total WBCs, elevated lymphocytes, anemia of chronic disease (microcytic hypochromic anemia with normal ferritin and TIBC), and elevated CRP.
- The TSH may be elevated, depressed, or normal, depending on the phase of the thyroiditis. Thyroid antibodies are negative in viral subacute thyroiditis, unless there is overlapping Hashimoto's or autoimmune thyroid response.
- A radioactive iodine scan will show almost no concentration of radioiodine in the thyroid gland because the cells are temporarily injured during the inflammatory response.
- The conventional treatment for thyroiditis typically involves aspirin or ibuprofen to decrease inflammation and pain. If no improvement occurs within a week, prednisone may be given in a dose of 40-60 mg daily, tapered to complete discontinuation over 4-6 weeks.
- If the thyroid is severely destroyed after the subacute process, thyroid hormone replacement therapy may be provided.

Suppurative Thyroiditis

- It is an extremely rare form of thyroiditis caused by bacterial infection (especially *Streptococcus pyrogens*, *Streptococcus aureus*, or *Streptococcus pneumoniae*).
- It is rare because antibiotics are commonly used, and bacterial infections are treated before they spread to the thyroid gland.
- It is also rare because the thyroid gland is generally resistant to infection because of its rich blood supply, lymphatic drainage, high iodine and hydrogen peroxide content, and encapsulation.
- The infection causes pus to collect and form an abscess in the thyroid gland. This may present clinically as an acute unilateral painful mass in their neck with erythema of the skin overlying the exquisitely tender thyroid. Fever, dysphagia, and dysphonia also are present.
- Laboratory testing typically demonstrates elevated ESR, CRP, WBCs, and thyroglobulin. The TSH may be normal, depressed, or elevated, depending upon the degree of thyroid destruction produced by the infection. Typically, the TSH is normal.
- Thyroid antibodies are normal, unless there are overlapping autoimmune reactions or Hashimoto's.
- Radioactive iodine scan will show almost no concentration of radioiodine by the thyroid gland because the cells are temporarily injured during the inflammatory response.
- Fine-needle aspiration of the lesion with gram stain in culture is the most useful diagnostic test.
- Conventional treatment is parenteral (IV) antibiotics and possible surgical drainage of the infected thyroid abscess.

Radiation-Induced Thyroiditis

- It is typically seen in hyperthyroid patients who are being treated by radioactive iodine therapy to manage of hyperthyroidism.
- The most important clinical differential diagnostic factor for this pattern is the development of pain in the thyroid and increased hyperthyroid symptoms 5-10 days after therapy.

- 1% of patients being treated with radioactive iodine develop radiation thyroiditis 5-10 days after the therapy.
- Radiation-induced thyroiditis may also develop from radiation therapy used to treat lymphoma of the head and neck.
- The major risk factors for developing thyroid damage after external irradiation are high-dose irradiation, younger age, female gender, and preexisting hypothyroidism.
- It is usually caused by too much substance leading to rapid destruction of the thyroid parenchyma resulting in further inflammation of the thyroid gland.
- It is important to remember that hyperthyroidism from an autoimmune mechanism is painless; therefore, if pain develops after radioactive iodine therapy, there is a radiation- induced thyroiditis.
- The gland eventually undergoes extensive fibrosis in approximately 6-18 weeks, and the patient may develop lifelong hypothyroidism from the destructive process.
- Conventional treatment for radiation-induced thyroiditis includes NSAIDs, prednisone, and beta-blockers to minimize the impact of increased thyroid hormone influences on peripheral basal metabolic rate.

Trauma-Induced Thyroiditis

- It is a rare condition, but may occur with significant trauma to the thyroid gland, such as being punched in the neck, being hit by a baseball to the neck, having a seatbelt restrain the body against the thyroid during a car accident, etc.
- The thyroiditis is self-limiting, and the history with obvious signs of bruising allows the diagnosis to be made.

Postpartum Thyroiditis

- Occurs in approximately 5%-7% of women who give birth, within two to four months after delivery.
- It is a result of an autoimmune process.
- It should be noted that women with euthyroidism who have TPO antibodies have a 25% risk for developing postpartum thyroiditis.[192]
- Additionally, patients with postpartum thyroiditis who have elevated thyroid antibodies have a 70% risk of recurrence following a subsequent pregnancy.[194]
- It is a common etiology for postpartum depression.
- During the third trimester of pregnancy, a female is TH-2 dominant and becomes TH-1 dominant postpartum. These immunological polar shifts, in combination with certain genotypes, may lead to the development of postpartum thyroiditis. It presents as a painless, small, nontender, firm goiter. Typically, the patient notices either an enlarged thyroid and/or symptoms of hyperthyroidism that include: increased anxiety, restlessness, insomnia, weight loss, and difficulty concentrating.
- Hyperthyroidism usually occurs 2-10 months postpartum, most commonly at three months, with recovery taking place over the next two to three months after the onset.
- Hypothyroidism occurs four to eight months postpartum, most commonly at six months. Most patients (80%) have normal thyroid function at one year. However, 30%-50% of patients develop permanent hypothyroidism within nine years.[192-193]
- Elevated levels of thyroid antibodies are found in 80% of patients, but the erythrocyte sedimentation rate is typically normal.
- It is important to distinguish painless thyroiditis from Graves' disease occurring in the postpartum period. The presence of an audible bruit over the gland, exophthalmos, hypervascularity with increased blood flow seen on Doppler ultrasonography, thyroid-stimulating immunoglobulins in the serum, and high RAIU are characteristic of Graves' disease but not postpartum thyroiditis.
- Thyroid uptake and scan should not be performed in women who are breastfeeding.
- It should be noted that women with euthyroidism who have TPO antibodies have a 25% risk for developing postpartum thyroiditis.[192]
- Additionally, patients with postpartum thyroiditis who have elevated thyroid antibodies have a 70% risk of recurrence following a subsequent pregnancy.[194]

Subacute Lymphocytic Thyroiditis

- It is of an autoimmune origin and usually presents with a small goiter (50% of cases), hyperthyroid responses, and is painless.
- The thyroid antibodies are positive in 50% of the cases.
- It clinically mimics a hyperthyroid Hashimoto's thyroiditis, especially when antibodies are present, but the histopathology of the condition is different.
- In subacute lymphocytic thyroiditis, no fibrosis occurs. The condition can only be differentially diagnosed from Hashimoto's thyroiditis that is active by FNA biopsy.
- The condition is very rare compared to Hashimoto's thyroiditis.

- It is differentiated from subacute thyroiditis (not lymphocytic) by the absence of thyroid pain and tenderness.
- It is differentiated from Graves' disease by the lack of thyroid thrill or bruit, no ophthalmopathy, no pretibial myxedema, no TSI elevations, by a low or absent rather than elevated RAIU, and FNA biopsy.
- The hyperthyroid stage averages three to four months. Eventually, these individuals will develop hypothyroidism from less-active autoimmune response in subsequent years.

Drug-Induced Thyroiditis

- Certain medications in the presence of autoimmune thyroid physiology may promote thyroiditis due to their stimulatory impact on up-regulating an autoimmune response.
- Amiodarone is rich in iodine content (37.3% by weight) and resembles thyroid hormones with its ability to bind to nuclear thyroid receptors. It may promote increased thyroid activity by increasing the thyroid nuclear response, and it may also feed a source of iodine to an autoimmune hyperactive response.
- Interferon (Infergen®; Intron A®, Roferon®-A combination therapy) and interleukin-2 are typically used to simulate the immune system in cases such as hepatitis C, cancer, HIV, etc. Unfortunately, it may promote an increased TH-1 cell-mediated response and increase the immune attack on the thyroid. The majority of autoimmune Hashimoto's are TH-1 dominant.
- Long-term treatment of lithium for manic-depressives and depression has been shown to promote both hypothyroidism and, less commonly, hyperthyroidism. Lithium may act as a goitrogen and block iodine uptake leading to hypothyroidism. However, lithium's blocking of iodine uptake may increase the metabolic activity of the thyroid gland and promote a hyperactive thyroiditis in the presence of a preexisting autoimmune thyroid disorder.
- These drugs may cause destructive thyroiditis in the presence of thyroid autoimmune diseases, as evidenced by elevated thyroid antibodies.
- It should be noted that the thyroid abnormalities usually resolve with discontinuation of the responsible drug.

Multinodular Goiter

- Multinodular goiter is a structurally and functionally heterogeneous thyroid enlargement, most often caused by malnutrition, medication, defects in hormone synthesis, and growth-stimulating antibodies.
- Palpable thyroid nodules may be detected in 0.8%-1.5% of men and 5.3-6.4% of women. Most of these nodules are benign colloid nodules composed of enlarged follicles containing abundant colloid.
- Palpation of the thyroid gland may reveal multiple nodules that all have similar consistency in relation to size and mobility.
- Multinodular thyroid may be toxic or nontoxic, but it is typically nontoxic.
- Indications for treatment of nontoxic nodules are compression of the trachea or esophagus, venous-outflow obstruction, growth of the goiter, neck discomfort, and cosmetic issues.

Diffuse Goiter

- Goiter is a diffuse enlargement of the thyroid gland. It is termed as toxic or nontoxic, depending on the resultant changes in thyroid hormone levels.
- Nontoxic goiter is a simple enlargement of the thyroid gland without any increased secretion of thyroid hormones.
- Nontoxic goiter can be endemic, most commonly due to a regional deficiency of iodine, or it can be sporadic.
- Sporadic goiter is typically a result of numerous etiologies including pregnancy, Hashimoto's, goitrogen-induced, drug-induced, environmentally induced, or any mechanism that may raise TSH levels.
- Goiter may also be a result of thyroiditis promoting inflammatory enlargement of the gland.
- Pregnancy may also promote nontoxic goiter because elevated human chorionic gonadotropin (hCG) can promote enlargement of the thyroid gland.
- Most clinical cases of nontoxic goiter today are sporadic and are typically not caused by simple iodine deficiency but rather by dysfunction in thyroid physiology leading to increased stimulation of TSH.
- Chronic TSH stimulation promotes enlargement of the thyroid.
- On the other hand, toxic goiter is an enlargement of the thyroid gland that produces pathologically excess amounts of thyroid hormones. Toxic goiter occurs in three forms, namely Graves' disease, toxic adenoma, and toxic nodular goiter. Toxic goiter leads to disorders promoted by increased basal metabolic rate from excess thyroid hormone release, such as cardiac arrhythmias, loss of weight, weakness, hyperkinesia, etc.

Please see registered trademark ownership in Appendix 1.

- Once a diffuse enlargement of the thyroid gland is noted (goiter), simple observations in the patient's concomitant symptoms can help differentiate between toxic and nontoxic goiter.
- If the patient with a goiter has symptoms associated with increased basal metabolic activity, then toxic goiter is likely.
- If the patient with goiter has symptoms associated with low basal metabolic activity, then nontoxic goiter is likely.
- It should be noted that many patients with nontoxic goiter will not present with any thyroid symptoms.
- Follow-up laboratory testing to assess TSH, T4, T3, and thyroid antibodies should be conducted to further evaluate the goiter.
- Additionally, thyroid ultrasound, RAIU scan, or FNA biopsy may be conducted to confirm that the goiter is not a result of malignant tissue growth.
- Typically, multinodular (not solitary) or diffuse enlargements of the thyroid are not due to malignancy.
- The incidence of malignancy within a multinodular goiter is usually significantly less than 5%.

Nontoxic Goiter

- The term nontoxic goiter refers to enlargement of the thyroid that is not associated with overproduction of thyroid hormones or malignancy.
- The thyroid becomes enlarged due to chronic stimulation of TSH.
- Although the most common cause is iodine deficiency worldwide, it is not the most common cause in the United States.
- Goiter development in the U.S. is usually due to a defect in normal thyroid synthesis.
- If the goiter becomes very large, it may produce compressive symptoms in surrounding tissue, such as the trachea and esophagus.
- Symptoms include coughing, waking up from sleep with feelings of difficulty breathing, and the sensation that food is getting stuck in the upper throat.
- Nontoxic goiter may be treated if compression takes place or for cosmetic reasons.
- Treatment for goiters should attempt to find the mechanism if TSH is elevated, such as iodine deficiency, drug-thyroid interaction, etc.
- In most instances, conventional endocrinologists will supply thyroid hormones to stabilize the size of the goiter. Thyroid hormones often will not cause the size of the goiter to decrease, but it will usually keep it from growing any larger.
- If the mechanism for the goiter cannot be identified and thyroid hormones are ineffective to reduce the size of the goiter, surgery may be recommended.

Toxic Goiter

- In toxic goiter, the thyroid gland is diffusely hyperplastic and excessively overproduces thyroid hormones. This results in accelerated metabolism in most body tissues.
- Toxic goiter occurs in three forms, namely Graves' disease, toxic adenoma, and toxic nodular goiter.
- The most common source (85%) is Graves' disease that can easily be identified by TSH antibodies/ immunoglobulins (TSI).
- In some cases of Graves' disease there are genetic hypersensitivity reactions to an autoantibody of IgG type class that acts on the receptors for TSH, leading to enlargement of the thyroid epithelium and the development of toxic goiter. (In nontoxic goiter, TSH stimulation promotes enlargement. In Graves' toxic goiter the TSH antibody promotes thyroid tissue enlargement.)
- If an autoimmune mechanism is not the etiology of toxic goiter, then toxic adenoma or toxic nodular tissue is autonomously promoting thyroid hormone secretion.
- Anatomically, they present as an isolated nodule more than a diffuse enlargement of the thyroid.
- Autonomous hyperactivity is conferred by somatic mutations of TSH hormones or TSH receptors in a majority of toxic adenomas and toxic nodules.
- Toxic nodular and adenomatous goiters appear to develop more frequently in areas of endemic iodine deficiency.
- Iodine deficiency promotes thyroid cell hyperplasia and increases thyroid cell replication, which then leads to greater predisposition of the development of TSH hormone and receptor mutations, leading to toxic nodule.
- However, once toxic nodule or adenoma develop, iodine supplementation may promote greater thyroid hyperactivity (Jod-Basedow effect).

Thyroglossal Cysts

- A thyroglossal cyst is a fibrous cyst that forms from persistent thyroglossal duct during embryological development for a small portion of the population and presents as a midline neck lump.
- Thyroglossal duct cysts most often present with a palpable asymptomatic midline neck mass below the level of the hyoid bone.

- The mass on the neck moves during swallowing.
- Diagnostically, a thyroglossal cyst will move upwards with protrusion of the tongue.
- Some patients will have neck or throat pain, or dysphagia. The persistent duct or sinus can promote oral secretion, which may cause cysts to become infected.
- Up to half of thyroglossal cysts are not diagnosed until adult life.
- The tract can lie dormant for years or even decades until some kind of stimulus leads to cystic dilation.
- Infection can sometimes cause the transient appearance of a mass or enlargement of the cyst, at times with periodic recurrences.
- Spontaneous drainage may also occur. Treatment for a thyroglossal cyst is surgical resection, often requiring concomitant removal of the midsection of the hyoid bone.
- Although generally benign, the cyst will be removed if the patient exhibits difficulty in breathing or swallowing, or if the cyst is infected.
- Even if the symptoms are not present, the cyst may be removed to eliminate the chance of infection or development of a carcinoma, or for cosmetic reasons if there is unsightly protrusion from the neck.
- If the mass does not move with tongue protrusion, then it is a thyroid enlargement and palpation of the thyroid can further subdivide the mass into a diffuse multi-nodular mass, or a solitary nodule.

Chondroma

- A chondroma is a benign cartilaginous tumor that is encapsulated within the lobular growing patterns of cartilage. It presents as a mass over the thyroid that does not move with swallowing.

Dermoid Cyst

- A dermoid cyst is a benign cystic teratoma that presents as a mass over the thyroid that does not move with swallowing.

Papillary Thyroid Cancer

- Thyroid cancers typically present as a dominant solitary nodule. However, 95% of solitary thyroid nodules are benign.
- Thyroid cancer does not influence thyroid hormone synthesis; therefore, thyroid panels are normal. The exception is metastasis of cancer into the thyroid hormone–producing cells leading to thyroid hyperfunction.
- Most common of all thyroid cancers
- Peak onset ages 30-50
- Prognosis directly related to size (less than ½ inch has a good prognosis)
- Accounts for 85% of thyroid cancers due to radiation exposure
- Spread to lymph nodes of the neck present in more than 50% of cases
- Distant spread to lung or bones is very uncommon
- Overall cure rate very high (nearly 100% for small lesions in young patients)

Follicular Thyroid Cancer

- Thyroid cancers typically present as a dominant solitary nodule. However, 95% of solitary thyroid nodules are benign
- Thyroid cancer does not influence thyroid hormone synthesis; therefore, thyroid panels are normal. The exception is metastasis of cancer into the thyroid-hormone producing cells leading to thyroid hyperfunction.
- Peak onset ages 40-60
- Females more common than males by a 3:1 ratio
- Prognosis directly related to tumor size (less than 3/8 inch has a good prognosis)
- Rarely associated with radiation exposure
- Spread to lymph nodes is uncommon (10%)
- Invasion into vascular structures (veins and arteries) within the thyroid gland is common
- Distant spread to lungs and bones is uncommon, but more common than with papillary cancer
- Overall cure rate high (near 95% for small lesions in young patients), decreases with advanced age.

Medullary Thyroid Cancer

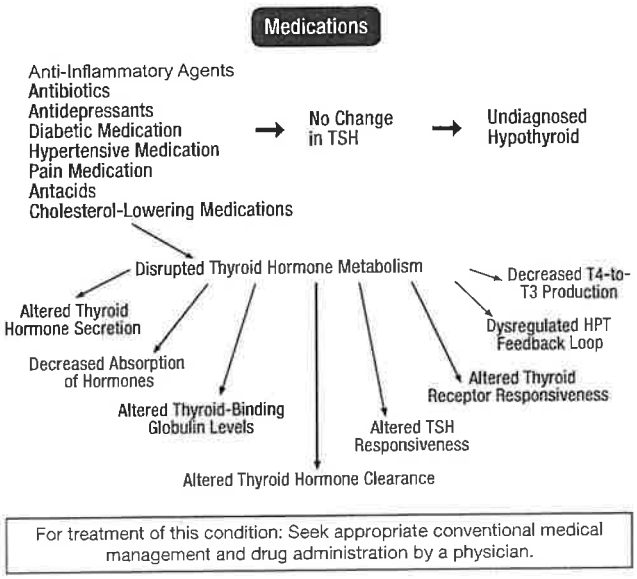
- Thyroid cancers typically present as a dominant solitary nodule. However, 95% of solitary thyroid nodules are benign.
- Thyroid cancer does not influence thyroid hormone synthesis therefore thyroid panels are normal. The exception is metastasis of cancer into the thyroid hormone producing cells leading to thyroid hyperfunction.
- Females more common than males (except for inherited cancers)
- Regional metastasis (spread to neck lymph nodes) occurs early.
- Spread to distant organs occurs late and can be to the liver, bone, brain, and adrenal medulla.

- Not associated with radiation exposure
- Poor prognostic factors include age older than 50, male, metastasis, and when seen in patients with other endocrine tumors due to MEN (multiple endocrine neoplasia)-II syndrome
- Residual disease (following surgery) or reoccurrence can be detected by measuring calcitonin every four months for the first few years than every six months after that.

Anaplastic Thyroid Cancer

- Thyroid cancers typically present as a dominant solitary nodule. However, 95% of solitary thyroid nodules are benign.
- Thyroid cancer does not influence thyroid hormone synthesis; therefore, thyroid panels are normal. The exception is metastasis of cancer into the thyroid hormone–producing cells, leading to thyroid hyperfunction.
- Peak onset age 65 or older
- Very rare in young patients
- Males more common than females by a 2:1 ration
- Typically presents as a rapidly growing neck mass
- Can occur many years after radiation exposure
- Spread to lymph nodes of the neck present in more than 90% of cases
- Distant spread to lungs and bones is very common even when first diagnosed
- Overall cure rate is very low

Medications That Impact Thyroid Physiology



Anti-addiction agents

- Methadone: dolophine hydrochloride tablets, methadone HCl powder, methadone dispersible tablets, methadone oral tablets

These anti-addiction drugs alter T4 and T3 transport in serum by increasing TBG concentrations. This would reflect as a thyroid panel with depressed T3 uptake and variations between total and free T4 and T3 levels.

Antibiotics

- Ethionamide: Trecator®-SC tablets
- Ketoconazole: Nizoral® tablets, ketoconazole tablets
- Para-aminosalicylic acid: Paser® granules
- Sulfonamide drugs: Bactrim® tablets, Septra® tablets
- Rifampin: Rifadin® capsules, Rifater®

These antibiotic drugs have been suggested to potentially impact the absorption of thyroxine, as well as potentially have an antithyroid effect by decreasing thyroid gland hormone synthesis and secretion. This would reflect on a thyroid panel as a pattern consistent with hypothyroidism, as defined by an elevated TSH with low or normal T4 and T3 levels.

Anti-inflammatory agents

- Glucocorticoids: Celestone®, Decadron®, Depo-Medrol®, hydrocortisone tablets, Solu-Medrol® Sterile Powder
- Aspirin: Darvon Compound, Ecotrin® enteric aspirin, Excedrin®, Fiorinal® halfprin tablets, Norgesic™ tablets, Percodan® tablets, Robaxisal® tablets, Soma® Compound, gelpirin tablets, propoxyphene compound, Roxiprin® tablets

These anti-inflammatory medications have been shown to suppress TSH release, leading to decreased thyroid gland activation. This would reflect as TSH below the functional range, but in the low end of normal laboratory range with potentially depressed T4 and/or T3.

Hyperthyroid medications

- Propylthiouracil: propylthiouracil tablets
- Methimazole: Tapazole® tablets, methimazole tablets

These hyperthyroid medications inhibit the enzyme TPO, which normally acts in thyroid hormone synthesis to add iodide to the tyrosine residues on the hormone precursor thyroglobulin, thus forming thyroxine. These drugs would only be used in obvious hyperthyroid cases and suppression of hyperactivity could be monitored with laboratory profiles by evaluating normalization of T4 and T3 levels.

Growth hormone–modulating agents

- Octreotide: Sandostatin®

These growth hormone–modulating agents suppress TSH release. They are typically used with acromegaly and TSH-secreting tumors.

Antinausea medications

- Metoclopramide: Reglan®, metoclopramide tablets

TSH secretion is controlled by the stimulatory action of hypothalamic TRH and the inhibition via central dopaminergic and somatostatinergic mechanisms as well as by a hypothalamic inhibitory action of thyroid hormones. Metoclopramide is a dopamine (D2) receptor blockade that can accelerate to elevate the concentration of serum TSH in hypothyroidism, which is accompanied by a progressive loss of dopaminergic tone, as demonstrated by a progressive increase in prolactin serum levels.

Diabetic medications

- Tolbutamide: Orinase®
- Chlorpropamide: Diabinese®

Please see registered trademark ownership in Appendix 1.

These diabetic medications have an antithyroid effect after prolonged therapy by causing a defect in the organic binding of iodine. The effects of this would present as a hypothyroid pattern of elevated TSH on lab tests. They also block binding of T4 to the carrier proteins and thus depress T4 concentrations.

Diuretics

- Furosemide: Lasix®, furosemide tablets

These diuretic medications are inhibitors of thyroid-hormone binding in serum, and high-dose treatment can lower total thyroxine and increase its free fraction.

Stimulants

- Amphetamine: Adderall® tablets

These stimulant medications increase the sensitivity of thyroid receptors to thyroid hormones and potentially increase the release of TSH. These reactions may be seen as increased TSH and T4, or the lab test may not demonstrate any change despite symptoms of increased metabolic activity due to the drug's influence on thyroid receptors directly.

Cholesterol-lowering medications

- Clofibrate: Atromid-S® capsules
- Cholestyramine: LoCholest® powder, Questran® Light, Prevalite®
- Colestipol: Colestid® tablets

These cholesterol-lowering medications decrease the absorption of T4 by disrupting enterohepatic circulation of bile acids by sequestering them and preventing their reabsorption from the gut. The effects of these drugs may demonstrate unchanging laboratory TSH and hypothyroid symptomatology, despite THR.

Arrhythmia medications

- Amiodarone: Cordarone® tablets, Pacerone® tablets
- Propranolol: Inderal® tablets, propranolol HCl Tablets
- Phentolamine: Regitine® vials

These arrhythmia medications decrease the conversion of T4 into T3 hormones by decreasing the activity of the enzyme 5'-deiodinase. These medications may cause a pattern of depressed T3 with normal T4 and TSH.

Hormone replacement

- Estrogens: estinyl tablets, Estrace® vaginal cream, Estratab® tablets, Menest® tablets, Ogen® tablets, Premarin® tablets, Vagifem® tablets

These estrogen medications increase the production of thyroid-binding globulin and decrease the percentage of free-fraction hormones. They will present on laboratory testing as decreased T3 uptake, decreased free T4 and/or T3, with normal protein-bound levels of T4, normal protein-bound levels of T3, and normal TSH.

- Estrogen and progestin combinations: Activella® tablets, Brevicon® 28-day tablets, Demulen, Desogen® tablets, Estrostep® 21 tablets, Levora® tablets, Ovral® tablets, Mircette® tablets, Modicon® tablets, Necon® tablets, Ortho-Cyclen® tablets, Ortho-Novum® tablets, Ovcon® tablets, Premphase® tablets, Prempro® tablets, Tri-Norinyl®-28 tablets, Triphasil® tablets, Trivora® tablets, Zovia tablets

These estrogen medications increase the production of thyroid-binding globulin and decrease the percentage of free-fraction hormones. They will present on laboratory testing as decreased T3 uptake, decreased free T4 and/or T3, with normal protein-bound levels of T4, normal protein-bound levels of T3, and normal TSH.

- Anabolic steroids: Anadrol®-50 tablets, Oxandrin® tablets, Winstrol tablets

These anabolic medications decrease the production of thyroid-binding globulin and increase the percentage of free-fraction hormones. They will present on laboratory testing as increased T3 uptake, increased free T4 and/or T3, with normal protein-bound levels of T4, normal protein-bound levels of T3, and normal TSH.

- Androgens: Androderm® transdermal system, AndroGel®, Android® capsules, Delatestryl® injection, Testoderm® transdermal systems, Testred® capsules

These anabolic medications decrease the production of thyroid-binding globulin and increase the percentage of free-fraction hormones. They will present on laboratory testing as increased T3 uptake, increased free T4 and/or T3, with normal protein-bound levels of T4, normal protein-bound levels of T3, and normal TSH.

- Growth Hormones: Geref® for injection, Humatrope®, Nutropin®, Protropin®, Saizen® for injection

These growth hormone medications may promote the production of TSH, T4, and T3 levels.

Pain medications

- Morphine: Astramorph/PF injection, Duramorph injection, Kadian® capsules, MS Contin® tablets, MSIR oral capsules, Oramorph SR tablets, Roxanol™ 100 concentrated oral solution, morphine sulfate, OMS concentrate CII, RMS suppositories CII

Please see registered trademark ownership in Appendix 1.

These pain medications increase the production of thyroid-binding globulin and decrease the percentage of free-fraction hormones. They will present on laboratory testing as decreased T3 uptake, decreased free T4 and/or T3, with normal protein-bound levels of T4, normal protein-bound levels of T3, and normal TSH.

Antacids

- Aluminum hydroxide: Amphojel®, Maalox®, Mylanta®, Alu-cap™ capsules, alumina and magnesia oral suspension

These antacid medications decrease the absorption of T4. The effects of these drugs may demonstrate unchanging laboratory TSH and hypothyroid symptomatology, despite THR.

Psychoactive medications

- Lithium (bipolar drug): Eskalith® capsules, Eskalith® CR controlled release tablets, lithium carbonate capsules, Lithobid® slow-release tablets
  - These drugs potentially reduce thyroid gland secretion of thyroid hormones
- Chlorpromazine (antipsychotic): Thorazine®, chlorpromazine HCl
  - These drugs potentially reduce TSH secretion
- Haloperidol: Haldol®

These medications decrease the secretion of thyroid hormones from the thyroid gland. The laboratory presentation from these reactions would be identical to hypothyroidism and identified as depressed T4 and T3 levels with increased TSH.

Drugs That Influence Thyroid Metabolism

New England Journal of Medicine. 1995 Dec; 333(25):1688, Table 2.

Many commonly prescribed medications have adverse impacts on thyroid physiology. The clinician must become familiar with these drugs when examining patient profiles. Many times the drug-thyroid interaction is the cause of symptoms.

Drugs that decrease TSH secretion:

- **Dopamine agonists/antagonists:** These are typically antipsychotics and dopamine uptake inhibitor antidepressants. Excessive L-dopa therapy (levodopa, Sinemet®, Parcopa®, Atamet®, Stalevo®, Madopa®, Prolopa®, etc) for Parkinson's may also suppress TSH release.
- **Glucocorticoids:** These are typically anti-inflammatory medications used with inflammatory joint disease, degenerative joint diseases, Addison's, and autoimmune diseases.
- **Octreotide:** This drug is an octapeptide that mimics natural somatostatin, which suppresses the release of hormones and secretion of intestinal fluids. It is used for acromegaly, and for diarrhea in carcinoid syndrome and vasoactive intestinal peptide tumors.

Drugs that alter thyroid hormone secretion by decreasing thyroid gland secretion:

- **Lithium:** used to manage mood in bipolar disorders
- **Iodide:** used in microgram doses to support iodine-deficient hypothyroidism but does not alter thyroid hormone secretion until doses are greater than 6 milligrams
- **Aminoglutethimide:** an anti-steroid drug marketed under the tradename Cytadren®. It blocks the production of steroids derived from cholesterol and is clinically used in the treatment of Cushing's syndrome and metastatic breast cancer. It is also a drug of abuse by body builders.

Drugs that alter thyroid hormone secretion by increasing secretion:

- **Iodide:** used in microgram doses to support iodine-deficient hypothyroidism but does not alter thyroid hormone secretion until doses are greater than 6 milligrams
- **Amiodarone:** an antiarrhythmic agent (medication used for irregular heart beat) and used for various types of tachyarrhythmias (fast forms of irregular heart beat), both ventricular and supraventricular (atrial) arrhythmias

Drugs that decrease T4 absorption:

- **Colestipol:** a cholesterol-lowering bile acid. Sequestrants are a group of medications used to bind certain components of bile in the gastrointestinal tract. They disrupt the enterohepatic circulation of bile acids by sequestering them and preventing their reabsorption from the gut.
- **Cholestyramine:** a cholesterol-lowering bile acid. Sequestrants are a group of medications used to bind certain components of bile in the gastrointestinal tract. They disrupt the enterohepatic circulation of bile acids by sequestering them and preventing their reabsorption from the gut.
- **Aluminum hydroxide:** Pharmacologically, this compound is used as an antacid under names such as Alu-cap®, Aludrox®, or Pepsamar®. The hydroxide reacts with excess acid in the stomach, reducing its acidity. This decrease of acidity of the

Please see registered trademark ownership in Appendix 1.

stomach's contents may in turn help to relieve the symptoms's uclers, heartburn, or dyspepsia.

- **Ferrous sulfate:** used to fortify foods and to treat iron-deficiency anemia.
- **Sucralfate:** Used for treatment of peptic ulcers, sucralfate is a sucrose-aluminum complex that binds to the hydrochloric acid in the stomach and acts like an acid buffer.

Drugs that alter T4 and T3 transport in serum by increasing TBG concentrations:

- **Estrogens:** typically used for contraception in menstruating women and as hormone replacement therapy in menopausal women to help ameliorate hot flashes and other associated menopausal symptoms.
- **Tamoxifen:** an antagonist of the estrogen receptor in breast tissue. It has been the standard endocrine (anti-estrogen) therapy for hormone-positive early breast cancer.
- **Heroin:** a semi-synthetic opioid drug synthesized from morphine, a derivative of the opium poppy. As with other opioids, heroin is used as both a painkiller and a recreational drug, and has high potential for abuse.
- **Methadone:** a synthetic opioid, used medically as an analgesic, antitussive, and a maintenance anti-addictive for use in patients on opioids
- **Mitotane:** a medication used in the treatment of the rare adrenocortical carcinoma. It alters steroid peripheral metabolism, directly suppresses the adrenal cortex, and alters cortisone metabolism leading to hypercortisolism.
- **Fluorouracil:** a pyrimidine analog, which is used as a drug in the treatment of cancer. It belongs to the family of drugs called antimetabolites.

Drugs that alter T4 and T3 transport in serum by decreasing TBG concentrations:

- **Androgens:** used as treatments for hormone replacement therapy and certain cancers (breast)
- **Anabolic steroids:** used for illegal athletic performance, severe burns, and wasting diseases
- **Slow-release nicotinic acid:** used for hyperlipidemia or for niacin-flushing effects
- **Glucocorticoids:** used for autoimmune diseases, chronic inflammation, and immune suppression

Drugs that alter T4 and T3 transport by displacement from protein-binding sites:

- **Furosemide:** a loop diuretic used in the treatment of congestive heart failure and edema
- **Mefenamic Acid:** a nonsteroidal anti-inflammatory drug used to treat pain, including menstrual pain
- **Salicylates:** typically used as aspirin for pain and inflammation

Drugs that alter T4 and T3 metabolism by increasing hepatic metabolism:

- **Phenobarbital:** a barbiturate used as an anticonvulsant
- **Rifampin:** a bactericidal antibiotic drug
- **Phenytoin:** a commonly used antiepileptic
- **Carbamazepine:** an anticonvulsant and mood-stabilizing drug used primarily in the treatment of epilepsy and bipolar disorder, as well as trigeminal neuralgia

Drugs that decrease 5'-deiodinase activity:

- **Propylthiouracil:** a thioamide drug used to treat hyperthyroidism (including Graves' disease) by decreasing the amount of thyroid hormone produced by the thyroid gland
- **Amiodarone:** an antiarrhythmic agent (medication used for irregular heart beat) used for various types of tachyarrhythmias (fast forms of irregular heart beat), both ventricular and supraventricular (atrial) arrhythmias.

Beta-adrenergic-antagonist drugs: a class of drugs used for various indications, but particularly for the management of cardiac arrhythmias, cardioprotection after myocardial infarction, and hypertension

Summary Chart of Thyroid and Drug Interactions (Per U.S. National Library of Medicine [NIH])

Drug or Drug Class	Effect
Drugs that may reduce TSH secretion—the reduction is not sustained; therefore, hypothyroidism does not occur	
Dopamine/dopamine agonists, glucocorticoids, octreotide	Use of these agents may result in a transient reduction in TSH secretion when administered at the following doses: dopamine (≥1 mcg/kg/min), glucocorticoids (hydrocortisone ≥100 mg/day or equivalent), octreotide (>100 mcg/day).
Drugs that alter thyroid hormone secretion	
Drugs that may decrease thyroid hormone secretion, which may result in hypothyroidism	

Summary Chart of Thyroid and Drug Interactions (Per U.S. National Library of Medicine [NIH])

Drug or Drug Class	Effect
Aminoglutethimide, amiodarone, iodide (including iodine-containing radiographic contrast agents), lithium methimazole, propylthiouracil (PTU), sulfonamides, tolbutamide	Long-term lithium therapy can result in goiter in up to 50% of patients, and either subclinical or overt hypothyroidism, each in up to 20% of patients. The fetus, neonate, elderly, and euthyroid patients with underlying thyroid disease (eg, Hashimoto's thyroiditis or with Graves' disease previously treated with radioiodine or surgery) are among those individuals who are particularly susceptible to iodine-induced hypothyroidism. Oral cholecystographic agents and amiodarone are slowly excreted, producing more prolonged hypothyroidism than parenterally administered iodinated contrast agents. Long-term aminoglutethimide therapy may minimally decrease T4 and T3 levels and increase TSH, although all values remain within normal limits in most patients.
Drugs that may increase thyroid hormone secretion, which may result in hyperthyroidism	
Amiodarone, iodide (including iodine-containing radiographic contrast agents)	Iodide and drugs that contain pharmacologic amounts of iodide may cause hyperthyroidism in euthyroid patients with Graves' disease previously treated with antithyroid drugs or in euthyroid patients with thyroid autonomy (eg, multinodular goiter or hyperfunctioning thyroid adenoma). Hyperthyroidism may develop over several weeks and may persist for several months after therapy discontinuation. Amiodarone may induce hyperthyroidism by causing thyroiditis.
Drugs that may decrease T4 absorption, which may result in hypothyroidism	

Summary Chart of Thyroid and Drug Interactions (Per U.S. National Library of Medicine [NIH])

Drug or Drug Class	Effect
Antacids, aluminum & magnesium, hydroxides, simethicone, bile acid sequestrants, cholestyramine, colestipol, calcium carbonate, cation exchange resins, Kayexalate®, ferrous sulfate, orlistat, sucralfate	Concurrent use may reduce the efficacy of levothyroxine by binding and delaying or preventing absorption, potentially resulting in hypothyroidism. Calcium carbonate may form an insoluble chelate with levothyroxine, and ferrous sulfate likely forms a ferric-thyroxine complex. Administer levothyroxine at least four hours apart from these agents. Patients treated concomitantly with orlistat and levothyroxine should be monitored for changes in thyroid function.
Drugs that may alter T4 and T3 serum transport, but free T4 concentration remains normal; therefore, the patient remains euthyroid	
Drugs that may increase serum TBG concentration	Drugs that may decrease serum TBG concentration
Clofibrate, estrogen-containing oral contraceptives, estrogens (oral), heroin/methadone, 5-fluorouracil mitotane, tamoxifen	Androgens/anabolic steroids asparaginase, glucocorticoids, slow-release nicotinic acid
Drugs that may cause protein-binding site displacement	
Drugs that may alter T4 and T3 metabolism	
Drugs that may increase hepatic metabolism, which may result in hypothyroidism	
Carbamazepine, hydantoins phenobarbital, rifampin	Stimulation of hepatic microsomal drug-metabolizing enzyme activity may cause increased hepatic degradation of levothyroxine, resulting in increased levothyroxine requirements. Phenytoin and carbamazepine reduce serum protein binding of levothyroxine, and total T4 and T4 may be reduced by 20%-40%, but most patients have normal serum TSH levels and are clinically euthyroid.
Drugs that may decrease T4 5'-deiodinase activity	

Please see registered trademark ownership in Appendix 1.

Summary Chart of Thyroid and Drug Interactions (Per U.S. National Library of Medicine [NIH])

Drug or Drug Class	Effect
Amiodarone, beta-adrenergic antagonists (eg, propranolol >160 mg/day), glucocorticoids (eg, Dexamethasone ≥4 mg/day), propylthiouracil (PTU)	Administration of these enzyme inhibitors decreases the peripheral conversion of T4 to T3, leading to decreased T3 levels. However, serum T4 levels are usually normal but may occasionally be slightly increased. In patients treated with large doses of propranolol (>160 mg/day), T3 and T4 levels change slightly, TSH levels remain normal, and patients are clinically euthyroid. It should be noted that actions of particular beta-adrenergic antagonists may be impaired when the hypothyroid patient is converted to the euthyroid state. Short-term administration of large doses of glucocorticoids may decrease serum T3 concentrations by 30% with minimal change in serum T4 levels. However, long-term glucocorticoid therapy may result in slightly decreased T3 and T4 levels due to decreased TBG production (see above).
Miscellaneous	
Anticoagulants (oral), coumarin derivatives, indandione derivatives	Thyroid hormones appear to increase the catabolism of vitamin K–dependent clotting factors, thereby increasing the anticoagulant activity of oral anticoagulants. Concomitant use of these agents impairs the compensatory increases in clotting-factor synthesis. Prothrombin time should be carefully monitored in patients taking levothyroxine and oral anticoagulants and the dose of anticoagulant therapy adjusted accordingly.

Summary Chart of Thyroid and Drug Interactions (Per U.S. National Library of Medicine [NIH])

Drug or Drug Class	Effect
Antidepressants, tricyclics (eg, amitriptyline), tetracyclics (eg, maprotiline), selective serotonin reuptake inhibitors (SSRIs; eg, sertraline)	Concurrent use of tri/tetracyclic antidepressants and levothyroxine may increase the therapeutic and toxic effects of both drugs, possibly due to increased receptor sensitivity to catecholamines. Toxic effects may include increased risk of cardiac arrhythmias and CNS stimulation; onset of action of tricyclics may be accelerated. Administration of sertraline in patients stabilized on levothyroxine may result in increased levothyroxine requirements.
Antidiabetic agents, biguanides, meglitinides, sulfonylureas,thiazolidinediones, insulin	Addition of levothyroxine to antidiabetic or insulin therapy may result in increased antidiabetic agent or insulin requirements. Careful monitoring of diabetic control is recommended, especially when thyroid therapy is started, changed, or discontinued.
Cardiac glycosides	Serum digitalis glycoside levels may be reduced in hyperthyroidism, or when the hypothyroid patient is converted to the euthyroid state. Therapeutic effect of digitalis glycosides may be reduced.
Cytokines (eg, interferon-alpha, interleukin-2)	Therapy with interferon-α has been associated with the development of antithyroid microsomal antibodies in 20% of patients, and some have transient hypothyroidism, hyperthyroidism, or both. Patients who have antithyroid antibodies before treatment are at higher risk for thyroid dysfunction during treatment. Interleukin-2 has been associated with transient painless thyroiditis in 20% of patients. Interferon-beta and -gamma have not been reported to cause thyroid dysfunction.

Summary Chart of Thyroid and Drug Interactions (Per U.S. National Library of Medicine [NIH])

Drug or Drug Class	Effect
Growth hormones (eg, somatremsomatropin)	Excessive use of thyroid hormones with growth hormones may accelerate epiphyseal closure. However, untreated hypothyroidism may interfere with growth response to growth hormone.
Ketamine	Concurrent use may produce marked hypertension and tachycardia; cautious administration to patients receiving thyroid hormone therapy is recommended.
Methylxanthine, bronchodilators (eg, Theophylline)	Decreased theophylline clearance may occur in hypothyroid patients; clearance returns to normal when the euthyroid state is achieved.
Radiographic agents	Thyroid hormones may reduce the uptake of iodine-123, iodine-131, and technetium-99.
Sympathomimetics	Concurrent use may increase the effects of sympathomimetics or thyroid hormone. Thyroid hormones may increase the risk of coronary insufficiency when sympathomimetic agents are administered to patients with coronary artery disease.
Chloral hydrate, diazepam, ethionamide, lovastatin, metoclopramide, 6-mercaptopurine nitroprusside, para-aminosalicylate sodium, perphenazine, resorcinol (excessive topical use), thiazide, diuretics	These agents have been associated with thyroid hormone and/or TSH level alterations by various mechanisms.

Where a condition is identified or a lab result is outside the normal range: Patient should seek appropriate conventional medical management and drug administration by a physician.

Case Examples

Case 16

A patient had recent dental work, developed an infection, and was placed on a heavy dose of antibiotics. She had lab work done for her thyroid during this time. What do you expect as a lab result possibility?

Case 17

A patient has been taking oral contraceptives for the past three years. She has had hypothyroid symptoms the past year, but thyroid evaluations completed by her HMO doctor are all normal. What do you expect as a lab result possibility?

Case 18

A patient has severe rheumatoid arthritis and Graves' disease, which led to a thyroidectomy from thyrotoxicosis several years ago. She is currently taking Synthroid® and hydrocortisone. What influence will these medications have on a thyroid panel that assesses her physiology?

Case 19

A patient has been on hormone-replacement therapy for the past five years. She was initially placed on the hormones to manage hot flashes during perimenopause. She was told to stay on the hormones to prevent osteoporosis after perimenopause. Last year, she developed a breast mass that was identified as estrogen-positive malignant breast cancer. She is currently taking tamoxifen. What potential influence will it have on her lab panel?

Case 20

A patient complains of chronic depression and mood disorders. She has all the symptoms of hypothyroidism, but TSH measured on two occasions in the past five years indicated normal TSH levels. She was diagnosed with bipolar disorder last year and has been taking lithium. What mechanisms must you consider clinically?

Case 21

A patient has a history of anxiety attacks, depression, and mental fatigue. She has been taking levothyroxine and occasional Xanax® when her anxiety flares up. She states that she has noticed when she eats too many carbohydrates, her anxiety gets worse. What mechanisms must you consider?

Case 22

A patient with a history of hypothyroidism for 10 years has been taking Armour® at a consistent dosage until the past two years. During the past two years, she has been feeling much worse and has had to increase her dosage several times. She has also been having

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severe vertigo and nausea the past six months and has been taking metoclopramide (antinausea drug) for her symptoms. What mechanisms must be considered?

Case 23

A patient with hypothyroidism who has been on levothyroxine for the past 15 years was recently diagnosed with gastric reflux disorder and placed on antacid medications. She later visited her nutritionist and was placed on iron, calcium, and B12 to compensate for her inability to absorb these nutrients. She has recently noticed weight gain and hair loss. What mechanisms must be considered?

Case 24

The parents of a heroin-addicted 16-year-old boy are referred to your office for alternative medicine support. The teenager has just finished 18 months of rehab, and the parents now want you to improve his nutritional status and health. The child has been home for several months. You ran a blood chemistry panel that includes a thyroid panel. You see that his T3 uptake is depressed. What are some possible mechanisms?

Thyroid Hormone Replacement

Thyroid hormone replacement may be warranted when a person has become hypothyroid due to such conditions as Hashimoto's thyroiditis, thyroidectomy, or genetic disorders. However, it is first always important to find the mechanism for the thyroid demise clinically. For example, a person with Hashimoto's will typically have other autoimmune diseases that are unidentified. A person with Coxsackie-based viral thyroiditis may end up with chronic immune dysregulation due to the nature of chronic virus. A person with thyroid cancer may be sensitive to environmental factors that are promoting faulty cellular apoptosis. The thyroid is a very sensitive gland, and demise of the gland is typically a red flag that something in the patient's health needs to be investigated even after thyroid replacement therapy is provided.

If a patient develops a hypothyroid condition related to loss of thyroid hormone–producing cells as listed above, thyroid hormone replacement therapy must be strongly considered. If a person truly loses their ability to make thyroid hormones and it is ignored, serious consequences may occur. These include serious risk for osteoporosis and neurodegeneration, infertility, accelerated aging, chronic fatigue, depression, poor mental function, goiter development, etc.

In addition to situations in which thyroid hormones are obviously warranted, there are instances in which thyroid hormone therapy is used in a controversial manner. These instances include thyroid hormone use for easier weight loss, antiaging benefits, fertility, chronic fatigue syndrome, etc. It is not the goal of this section to justify what a practitioner or patient should do. In this section of the manual, a simple review of different types of thyroid hormone replacements will be provided.

Thyroid hormone replacement can be separated into several categories that include synthetic hormones, bioidentical hormones, T4-only hormone, T3-only hormone, or a combination of T3 and T4 hormones. The most commonly prescribed hormone is Synthroid®, which is isolated synthetic T4. Armour® is another popular thyroid hormone replacement, which is bioidentical T3 and T4. Several other combinations of hormones are available today and will be discussed. Before we review the most commonly used thyroid hormones today, let's review the history of thyroid hormone replacement.

History of Thyroid Hormone Replacement

In ancient cultures, when goiter was identified, the village healers often used thyroid gland consumption of cattle to support the patient. In the early 1890s, medical textbooks provided recipes for preparing animal thyroid glands as minced thyroid. In the 1940s, the use of dried, desiccated, animal thyroid gland was used for thyroid hormone replacement. This was a breakthrough for patients who suffered from thyroid disorders; however, each batch of tissue had different amounts of T3 and T4. The patient never knew what they were getting from dose to dose, which created problems of over- and under-dosing of hormones.

In 1955, Synthroid® was developed by Abbott Laboratories. It was the first major isolated synthetic standardized dose of T4. It was well accepted because it provided an exact dose of thyroid hormones, which could not be accomplished with desiccated thyroid hormones. At that time, T4 was used exclusively because everyone believed that the thyroid could always make adequate T3 (the active form) from precursor T4. Additionally, T3 therapy may be too powerful of a hormone and could cause palpitations and increased metabolic rate activity. Therefore, the past and the current standard for thyroid hormone replacement therapy is Synthroid®. Since Abbott developed Synthroid® in the 1950s, it has spent a lot of money on marketing to convince doctors that it is the only accurately dosed medication available, and that generic brands and bioidentical hormones cannot be trusted for their accuracy. This is still the current view of many healthcare professionals who are out of touch with the literature and engrossed in Synthroid® marketing.

Several other pharmaceutical companies have produced both generic and brand name T4 hormones since the introduction of Synthroid® (Levothroid®, Levoxyl®, Euthyrox®, and levothyroxine). Not that long ago, all of these drugs, despite their abundant prescriptions and usage, were not FDA approved. Thyroid hormone replacements had been grandfathered in as being the same category of drugs as natural thyroid hormones derived from desiccated thyroid hormones. These hormones had been in use since the early 1900s, prior to the formation of the FDA. In the mid 1990s, several

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studies found that the doses of all T4-based hormones had demonstrated problems of stability, consistency, and potency. Finally, in 1997 the FDA called for all T4 drugs to be classified as new drugs and required all of them to submit new drug applications. Their deadline for approval was August 14, 2000. Eventually, the T4-based medications were forced to improve their quality to make them comply with FDA guidelines in subsequent years. Synthroid®, the most commonly prescribed thyroid hormone for the past 50 years, received FDA approval on July 24, 2002.

Since the inception of Synthroid® and synthetic T4 medications, two other major categories of thyroid hormones were developed. The first was synthetic T3 medications (or T4/T3), and the second was bioidentical hormones (T4/T3) that were consistent in their dosage. The most commonly used bioidentical T4/T3 thyroid hormone that is used in the country is Armour®, by Forest Pharmaceuticals. Armour® had been in the market even before Synthroid®, but doctors were being told that Armour® has no consistency in the amount of thyroid hormones per batch, and that the amount of T3 and T4 in each batch was inconsistent and could not be trusted. This is still an outdated belief held by many physicians. Armour® thyroid contains an exact dosage of 4.22:1 ratio of T4:T3. This can be accomplished with natural porcine sources because the batches of desiccated thyroid are mixed until the desired ratio is obtained.

Thyroid Replacement Hormones

Brand Name Thyroid Hormones

- **Synthroid®** – synthetic T4 only
- **Levothroid®** – synthetic T4 only
- **Levoxyl®** – synthetic T4 only
- **Unithroid®** – synthetic T4 only
- **Cytomel®** – synthetic T3 only
- **Thyrolar®** – synthetic fixed ratio of T3:T4
- **Armour®** – bioidentical fixed ratio of T3:T4 combination
- **Westhroid™** – bioidentical fixed ratio of T3:T4 combination
- **Nature-Throid®** – bioidentical fixed ratio of T3:T4 combination

Generic Thyroid Hormones

- **Levothyroxine** – synthetic T4 only
- **L-Thyroxine** – synthetic T4 only
- **Liothyronine** – synthetic T3 only
- **Liotrix** – synthetic fixed ratio of T3:T4 combination
- **Desiccated Thyroid** – bioidentical T3 and T4 combination

Synthetic T4 Hormones

- **Synthroid®** – brand name
- **Levothroid®** – brand name
- **Levoxyl®** – brand name
- **Unithroid®** – brand name
- **Levothyroxine** – generic
- **L-Thyroxine** – generic

Synthetic T3 Hormone

- **Cytomel®** – brand name
- **Liothyronine** – generic

Synthetic T3 and T4 Combination

- **Thyrolar®** – brand name
- **Liotrix** – generic

Bioidentical T3 and T4 Combination

- **Armour®** – brand name
- **Westhroid™** – brand name
- **Nature-Throid®** – brand name
- **Desiccated Thyroid** – generic

Synthroid®

Synthroid®: levothyroxine sodium tablets. It contains synthetic crystalline L-3,3',5,5'-tetraiodothyronine sodium salt. Its inactive ingredients are acacia, confectioner's sugar (contains corn starch), lactose monohydrate, magnesium stearate, povidone, talc, and food dyes, depending on tablet strength.

Note: Corn starch may be a cause of sensitivity for some, especially Hashimoto's patients who are gluten sensitive or have celiac disease. Acacia is a TH-2 stimulator and may cause immune responses with TH-2 dominant Hashimoto's patients. Lactose is a common sensitivity for many people.

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Synthroid® comes in different dosages with different additives listed below.	
Strength (µg)	Color additive(s)
25	FD&C Yellow No. 6 aluminum lake
50	None
75	FD&C Red No. 40 aluminum lake, FD&C Blue No. 2 aluminum lake
88	FD&C Blue No. 1 aluminum lake, FD&C Yellow No. 6 aluminum lake, D&C Yellow No. 10 aluminum lake
100	D&C Yellow No. 10 aluminum lake, FD&C Yellow No. 6 aluminum lake
112	D&C Red No. 27 & 30 aluminum lake
125	FD&C Yellow No. 6 aluminum lake, FD&C Red No. 40 aluminum lake, FD&C Blue No. 1 aluminum lake
137	FD&C Blue No. 1 aluminum lake
150	FD&C Blue No. 2 aluminum lake
175	FD&C Blue No. 1 aluminum lake, D&C Red No. 27 & 30 aluminum lake
200	FD&C Red No. 40 aluminum lake
300	D&C Yellow No. 10 aluminum lake, FD&C Yellow No. 6 aluminum lake, FD&C Blue No. 1 aluminum lake

Levothroid®

Levothroid®: levothyroxine sodium tablets. It contains synthetic crystalline L-3,3',5,5'-tetraiodothyronine sodium salt. Its inactive ingredients are microcrystalline cellulose, calcium phosphate dibasic, povidone and magnesium stearate, and food dyes, depending on tablet strength.

Levothroid® comes in different dosages with different additives listed below.	
Strength (µg)	Color additive(s)
25	FD&C Yellow No. 6 aluminum lake
50	None
75	FD&C Blue No. 2 aluminum lake, FD&C Red No. 40 aluminum lake
88	FD&C Yellow No. 6 aluminum lake, FD&C Blue No. 1 aluminum lake, D&C Yellow No. 10 aluminum lake
100	FD&C Yellow No. 6 aluminum lake, D&C Yellow No. 10 aluminum lake

Levothroid® comes in different dosages with different additives listed below.	
Strength (µg)	Color additive(s)
112	D&C Red No. 27 aluminum lake, D&C Red No. 30 aluminum lake
125	FD&C Blue No. 1 aluminum lake, FD&C Red No .40 aluminum lake, FD&C Yellow No. 6 aluminum lake
137	FD&C Blue No. 1 aluminum lake
150	FD&C Blue No. 2 aluminum lake
175	FD&C Blue No. 1 aluminum lake, D&C Red No. 30 aluminum lake, D&C Red No. 27 aluminum lake
200	FD&C Red No. 40 aluminum lake
300	FD&C Yellow No. 6 aluminum lake, FD&C Blue No. 1 aluminum lake, D&C Yellow No. 10 aluminum lake

Levoxyl®

Levoxyl®: levothyroxine sodium tablets. It contains synthetic crystalline L-3,3',5,5'-tetraiodothyronine sodium salt. Its inactive ingredients are microcrystalline cellulose, croscarmellose sodium and magnesium stearate, and dye, depending on tablet strength.

Levoxyl® comes in different dosages with different additives listed below.	
Strength (µg)	Color additive(s)
25	FD&C Yellow No. 6 aluminum lake
50	None
75	FD&C Blue No. 1 aluminum lake, D&C Red No. 30 aluminum lake
88	FD&C Yellow No. 6 aluminum lake, FD&C Blue No. 1 aluminum lake, D&C Yellow No. 10 aluminum lake
100	FD&C Yellow No. 6 aluminum lake, D&C Yellow No. 10 aluminum lake
112	FD&C Yellow No. 6 aluminum lake, FD&C Red No. 40 aluminum lake, D&C Red No. 30 aluminum lake
125	FD&C Red No. 40 aluminum lake, D&C Yellow No. 10 aluminum lake
137	FD&C Blue No. 1 aluminum lake
150	FD&C Blue No. 1 aluminum lake, D&C Red No. 30 aluminum lake
175	FD&C Blue No. 1 aluminum lake, D&C Yellow No. 10 aluminum lake

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Levoxyl® comes in different dosages with different additives listed below.	
Strength (µg)	Color additive(s)
200	D&C Red No. 30 aluminum lake, D&C Yellow No. 10 aluminum lake

Unithroid®

Unithroid®: levothyroxine sodium tablets. It contains synthetic crystalline L-3,3',5,5'-tetraiodothyronine sodium salt. Its inactive ingredients are colloidal silicon dioxide, lactose, magnesium stearate, microcrystalline cellulose, corn starch, acacia, sodium starch glycolate, and dyes, based on tablet strength.

Note: Corn starch may be a cause of sensitivity for some, especially Hashimoto's patients who are gluten sensitive or have celiac disease. Acacia is a TH-2 stimulator and may cause immune responses with TH-2 dominant Hashimoto's patients. Lactose is a common sensitivity for many people.

Unithroid® comes in different dosages with different additives listed below.	
Strength (µg)	Color Additive(s)
25	FD&C Yellow No. 6 aluminum lake
50	None
75	FD&C Red No. 40 aluminum lake, FD&C Blue No. 2 aluminum lake
88	D&C Yellow No. 10 aluminum lake, FD&C Yellow No. 6 aluminum lake, FD&C Blue No. 1 aluminum lake
100	D&C Yellow No. 10 aluminum lake, FD&C Yellow No. 6 aluminum lake
112	D&C Red No. 27 aluminum lake
125	FD&C Yellow No. 6 aluminum lake, FD&C Red No. 40 aluminum lake, FD&C Blue No. 1 aluminum lake
150	FD&C Blue No. 2 aluminum lake
175	FD&C Blue No. 1 aluminum lake, D&C Red No. 27 aluminum lake
200	FD&C Red No. 40 aluminum lake
300	D&C Yellow No. 10 aluminum lake, FD&C Yellow No. 6 aluminum lake, FD&C Blue No. 1 aluminum lake

Cytomel®

Cytomel®: a liothyronine sodium tablet. It contains isolated synthetic T3. Its inactive ingredients are calcium sulfate, gelatin, starch, stearic acid, sucrose, and talc.

Cytomel® tablets: 5 µg in bottles of 100; 25 µg in bottles of 100; and 50 µg in bottles of 100. They are all white tablets without dyes. The dosage is clarified by imprints on the tables that list:

- 5 µg tablet imprinted with SKF/D14
- 25 µg tablet imprinted with SKF/D16
- 50 µg tablet imprinted with SKF/D17

Note: Inactive ingredients, such as starch, may be a hidden source of gluten.

Thyrolar®

Thyrolar®: liotrix tablets. It contains triiodothyronine (T3 liothyronine) sodium and tetraiodothyronine (T4 levothyroxine) sodium. The inactive ingredients are calcium phosphate, colloidal silicon dioxide, corn starch, lactose, and magnesium stearate. The tablets also contain the following dyes:

Thyrolar® comes in different dosages with different additives listed below.	
Strength (µg)	Color Additive(s)
¼	FD&C Blue No. 1 and FD&C Red No. 40
½	FD&C Red No. 40 and D&C Yellow No. 10
1	FD&C Red No. 40
2	FD&C Blue No. 1, FD&C Red No. 40, and D&C Yellow No. 10
3	FD&C Red No. 40 and D&C Yellow No. 10

Note: Corn starch may be a cause of sensitivity for some, especially Hashimoto's patients who are gluten sensitive or have celiac disease. Lactose is a common sensitivity for many people.

Armour®

Armour®: a natural preparation derived from porcine thyroid glands and has a strong, characteristic odor. It provides 38 µg T4 and 9 µg T3 per grain of thyroid. The inactive ingredients are calcium stearate, dextrose, microcrystalline cellulose, sodium starch glycolate, and Opadry® white.

Armour® tablets are evenly colored, light tan, round tablets, with convex surfaces. One side is debossed

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